IES ACADEMY

IES Academy

LEARNER GUIDE

For

GenAI for Engineers

Conducted by

IES Academy

Version 2.3

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
2.3	18 October 2026	Lab 11 Agent Builder work-order agent; Lab 12 group mini capstone; Copilot interface, AI security and Agent Builder pages; corrected key concepts; built-in @chart-design demo; dashboard with KPI tiles and slicers; lean lab folders; step-by-step prompts page.	IES Academy
2.2	14 October 2026	Ten connected labs: account Personalisation, visible .Rules, create/apply Pareto and reporting Skills, custom interactive-dashboard Skill, filters and controlled refresh.	IES Academy
2.1	13 October 2026	Water-pipe service scenario; expanded coherent prompts, pure-text Copy buttons, linked PDFs and simple data/PDF labs.	IES Academy
1.0	29 July 2026	Previous cross-app scenario; retained in archive.	IES Academy
2.0	13 October 2026	Four maintenance topics; eight connected labs, custom Excel/PowerPoint skills, interactive dashboard and management presentation.	IES Academy

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Introduction

This masterclass follows Northstar Water Services, a fictional water-pipe maintenance and service company, from a dirty service-work-order export to a reviewed management decision pack. Work-order IDs, categories, sites, current assignments and status definitions stay consistent across every app. The purpose is to make evidence travel accurately through the workflow, while using Copilot for drafting, analysis, reusable instructions and visual communication.

Study each topic's concepts before its connected labs. Topic 2 has four labs; Topics 1, 3 and 4 have two each; Topic 5 has the agent lab and the group mini capstone. Each lab has its own instruction PDF and a lean data folder: the lab workbook, only the templates that lab uses, and a rejoin checkpoint. Copy-ready prompts are on the linked web page, step by step. Carry your own verified outputs forward. A supplied checkpoint allows rejoining but is not evidence that you completed the earlier task. The detailed procedures are here and in labs; the slides teach concepts and show outcomes.

Course Learning Outcomes

- LO1: Integrate Word, Excel, PowerPoint and Outlook Copilot in a coherent maintenance workflow with explicit source grounding and verified prompts.
- LO2: Distinguish account Personalisation, workbook .Rules and reusable Skills; configure, create, apply and verify Pareto and maintenance-reporting Skills using water-service evidence.
- LO3: Create and apply a custom interactive-dashboard Skill with verified measures, shared filter scope, editable charts and controlled refresh and rerun tests.
- LO4: Continue from the dashboard to a professional management presentation, with evidence-based headlines, source notes, rehearsal and a reviewed handoff draft.
- LO5: Build and test a grounded work-order agent with Agent Builder, then apply the full data-to-decision workflow to your own mock or sanitised data in a group capstone.

Before You Start — Preparation

What you need

- Assigned Microsoft 365 account with Copilot in Word, Excel, PowerPoint and Outlook; licence and tenant settings checked before class.
- Excel desktop Windows or Mac recommended; current apps, English display language for Excel custom skills, assigned OneDrive and AutoSave.
- Complete lab folder downloads and a personal work-genai output folder; all sample data is synthetic.
- Copilot skills access in both Excel and PowerPoint. If unavailable, the lab supplies a clearly labelled manual-instruction practice route.

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Verify your setup

Open one workbook and one Word/PowerPoint template; find each Copilot button. Confirm that the app can select the intended OneDrive reference file. Use the app-created custom skills folder rather than guessing a path. Keep Formulas > Calculation Options > Automatic. Never paste account passwords into a prompt.

Conventions used in every lab

- App-labelled prompts run in the named app; a prompt in one app is not automatically visible to another.
- Use Chat only for inspection, Plan for reviewed proposals, and Allow editing for authorised workbook changes when these modes are available.
- Log Word, Outlook, PowerPoint and web prompts manually in Prompt_Log.
- All comparisons use the same source table, reporting date and filter scope. Missing values remain UNKNOWN; zero closed denominator is N/A.

Topic 1 — Fundamentals of Microsoft 365 Copilot

One verified maintenance workflow across four apps

Key concepts

- Large language models — Understand your request and generate natural-language answers.
- Grounding — Answers anchored in your work data: emails, files, meetings.
- Permissions & privacy — Shows only data you can access; prompts do not train the models.
- Prompts — State the goal, context, source and expectations.

Concepts for Lab 1: The Microsoft 365 Copilot Interface, Grounding Across Microsoft 365, Work-Order Grain and Reporting Measures.

The Microsoft 365 Copilot Interface

Copilot in the browser and in each Office app shares one set of controls. Microsoft Copilot (copilot.cloud.microsoft) and the Copilot pane in Word, Excel, PowerPoint and Outlook offer a model selector, new chat and chat history, a + menu to reference files and choose skills, settings for Personalization and Manage plugins & skills, and in Excel a mode switch between Chat only, Plan and Allow editing. Agents & Skills in the left pane opens your agents and Agent Builder.

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Knowing where each control lives lets you choose how much reasoning, context and authority Copilot gets for a task. A new chat starts a clean context; a reference adds a specific file; a mode decides whether Copilot only answers, proposes a plan or edits the file. The green shield confirms enterprise data protection for work data.

How it works

- Pick the model and start a new chat.
- Reference the file or choose a skill.
- Choose Chat only, Plan or Allow editing.
- Review history, settings and skills.

Worked example

- Auto is the default model; Think deeper suits multi-step analysis.
- Use / or + to reference the OneDrive workbook in Word.
- Chat only for inspection, Plan before edits, Allow editing to apply.

Decision guide

Use when	Avoid when
You need to control context, reasoning depth or edit permissions.	You assume a chat in one app is visible in another.

Authoritative references

- https://learn.microsoft.com/en-us/microsoft-365/copilot/microsoft-365-copilot-overview
- https://support.microsoft.com/en-us/excel/copilot/copilot-in-excel-skills

Grounding Across Microsoft 365

Grounding gives an AI request the specific evidence it should use. A work-grounded Copilot experience can use permitted organisational content; a file-grounded request uses selected files or supplied text. Neither makes an inaccessible file available merely because its local path is typed into a prompt. Word, Excel, PowerPoint and Outlook have different current-document and file-reference experiences. Treat each app handoff as a new grounding task.

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A maintenance figure can be correct in Excel yet become wrong in an email when the source, reporting date or filter scope changes. Explicit grounding prevents unsupported owner assignments, stale reporting periods and invented approvals from entering the pack. A Copilot response is a draft or hypothesis until the underlying evidence has been checked.

How it works

- Identify the authoritative work-order table.
- Provide goal, context, source and expected output.
- Review against the named evidence.
- Pass a verified snapshot to the next app.

Worked example

- Excel analyses tblWorkOrders; Word receives the checked summary.
- PowerPoint receives the reviewed Word brief; Outlook receives the same checked values.
- An Excel chat transcript is not an automatic PowerPoint data source.

Decision guide

Use when	Avoid when
The source is accessible and the output has an explicit audience.	The prompt assumes access to another app chat or local path.

Authoritative references

- https://support.microsoft.com/en-us/microsoft-365-copilot/file-formats-supported-by-microsoft-365-copilot
- https://support.microsoft.com/en-gb/word/copilot/draft-and-add-content-with-copilot-in-word

Work-Order Grain and Reporting Measures

A row represents one work order, identified by Work_Order_ID. Site, job category, priority, status and current assignment are dimensions used to group or filter records. Measures are defined calculations. The source is a snapshot at 30 September 2026: current backlog excludes Closed and Cancelled; overdue backlog also has a Due_Date before that date. Closed orders are counted separately from jobs still in the queue.

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An exact duplicated export row changes counts and cost without representing new work. Normalising a category spelling must not change the business meaning or assign a missing owner. Named Excel tables and structured references let formulas expand when legitimate new rows are appended. Preserve the raw export so a cleaning decision can be traced.

How it works

- Define one ID per work order.
- Find exact repeated export records.
- Standardise labels without inventing facts.
- Reconcile cleaned counts and cost.

Worked example

- 122 raw rows contain 120 distinct work-order IDs.
- Total = Closed + Cancelled + Backlog after exact duplicate removal.
- Due on 30 September is not overdue at the 30 September snapshot.

Decision guide

Use when	Avoid when
You have consistent IDs and a fixed reporting date.	Duplicate IDs contain conflicting records or reporting scope is unknown.

Authoritative references

- https://support.microsoft.com/en-us/excel/create-and-share-a-dashboard-with-excel-and-microsoft-groups

Lab 1 · Excel to Word — Prepare Work Orders and Draft the Operations Brief

Learning outcome: LO1: integrate verified Word, Excel, PowerPoint and Outlook outputs.

Goal: You are Northstar Water Services' maintenance coordinator. A Q3 export contains repeated IDs and inconsistent category labels. You will clean a copy in Excel, reconcile its controls, and use the verified result to draft a Word operations brief. The same site, status and assignment definitions underpin every later report.

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You are Northstar Water Services' maintenance coordinator. A Q3 export contains repeated IDs and inconsistent category labels. You will clean a copy in Excel, reconcile its controls, and use the verified result to draft a Word operations brief. The same site, status and assignment definitions underpin every later report.

What you'll build

Northstar_Clean_Work_Orders.xlsx and Northstar_Operations_Brief.docx, with reconciled counts, reporting scope and a source register. (Tools: Microsoft 365 Copilot · Excel · Word · PowerPoint · Outlook · OneDrive.)

Prerequisites

- Microsoft 365 account with Copilot available in Word, Excel, PowerPoint and Outlook. Availability depends on the assigned licence and tenant settings; confirm access with the trainer before class.
- Desktop Excel on Windows or Mac is recommended; save working copies to your assigned OneDrive course folder with AutoSave on. Do not overwrite the supplied input.
- Download this whole lab folder. All files are synthetic; work-genai is your own learner output folder.

Step-by-step

1. Copy data/Northstar_Work_Orders.xlsx to your OneDrive work-genai folder as Northstar_Clean_Work_Orders.xlsx. Open it in Excel, set Formulas > Calculation Options > Automatic, and read About_This_Data and Control. The workbook is the deliberately dirty 122-row input; the checkpoint workbook is a clean reference and must not replace the cleaning exercise. Open Copilot from Home > Copilot.

```
EXCEL - CHAT ONLY:

Read About_This_Data and Control for Northstar Water Services. Explain what
one water-pipe service work order represents, the reporting date, the five
job categories and how current owners differ from proposed owners. Do not
edit or infer customer impacts.
```

2. Use Chat only to inspect the named table before making changes. Record row count and the two exact repeated IDs. A Work_Order_ID is the row grain; do not count repeated export records twice.

```
EXCEL - CHAT ONLY:

Inspect tblWorkOrders. Report row count, duplicate Work_Order_ID values,
category spelling variants and missing Assigned_To. Do not edit. Treat
2026-09-30 as the fixed reporting date.
```

3. Choose Plan if available. Read the proposed cleaning plan before using Allow editing. The duplicates are exact repeated export records: keep one per ID. Trim and standardise category labels to Leak Repair, Valve Repair, Pipe Replacement, Pump

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Service, Meter Service; preserve missing owners and all unique source values. Save a copy of the raw sheet before the change.

```
EXCEL - PLAN:

Plan to retain an untouched Raw_Export copy, remove only exact duplicate
records by Work_Order_ID, and standardise category spelling in
tblWorkOrders. Keep dates, statuses and assignments. Ask if two records
with one ID differ.
```

```
EXCEL - ALLOW EDITING:

Apply the reviewed plan. Keep one identical record per repeated ID. Do not
guess missing owners. Reconcile the cleaned table against Control.
```

4. Add a Checks sheet or use the supplied one. Ask for formulas, then inspect the table and formulas yourself. Overdue excludes Closed and Cancelled and uses strictly earlier than the fixed date, so a due-today job is not overdue. Record each verified value in the Evidence_Log sheet (claim, value, source, scope and as-at date).

```
EXCEL - ALLOW EDITING:

In Checks calculate total rows, closed, cancelled, backlog, overdue
backlog, unassigned backlog and total Cost_SGD for tblWorkOrders. Use
Control!B2 as reporting date. Define backlog as Status neither Closed nor
Cancelled. Add the identity total = closed + cancelled + backlog. Do not
change source data.
```

5. Open data/templates/Northstar_Operations_Brief_Template.docx in Word and save as Northstar_Operations_Brief.docx in OneDrive. In Copilot select the verified source file using Reference a file or /. If Word cannot use the workbook, paste your verified evidence table plus the definitions from the Control sheet into the document and select that text; do not imply Word reads the Excel chat.

```
WORD COPILOT:

Draft a one-page Northstar operations brief from the verified evidence
supplied here. Include reporting scope/date, backlog and overdue findings,
current assignment issues, proposed next actions and open questions. Keep
observed facts separate from proposals. Preserve numbers and definitions.
Use UNKNOWN for causes, approvals or capacity decisions not evidenced.
```

6. Compare every number and work-order ID in the Word brief to Checks and the evidence register. Add the OneDrive source link and all-filters scope. Log the Word prompt manually in Excel Prompt_Log because Excel cannot access the Word conversation. Save the cleaned workbook and brief for Lab 2.

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```
WORD COPILOT:

Audit this Northstar Water Services operations brief against the verified
evidence table I supply. Return a discrepancy list for counts, cost,
reporting scope, overdue definition and assignment claims. Identify missing
source links as UNKNOWN. Do not change facts or approve work.
```

Sample prompts

Purpose	Prompt
Scope	State actual source, report date and selected scope before acting.
Verify	Explain the source/formula checks and preserve unknowns and original records.

Test it

Check	Expected result
Clean table	120 unique Work_Order_ID rows; five canonical job categories.
Portfolio identity	120 = 48 Closed + 12 Cancelled + 60 backlog.
Evidence handoff	Overdue 50; unassigned backlog 6; total cost S$35,827.50. Word agrees with Excel.
No guessed owner	Blank assignments remain blank; proposals labelled unapproved.

Checkpoint and rejoin point

If you fell behind, use data/checkpoints/Northstar_Verified_Checkpoint.xlsx and data/checkpoints/Northstar_Verified_Snapshot.docx. Record checkpoint used in Prompt_Log; then perform this lab, rather than claiming the checkpoint as your output.

Troubleshooting

If this happens	Fix
Copilot cannot find the source	Use the app file picker to select the actual OneDrive file. A typed local path or a different app chat is not shared context. If needed copy only the verified snapshot into the current app.
A total disagrees with Control	Clear dashboard filters to All, check table extent and duplicate IDs, recalculate, then compare definitions and reporting date. Reconcile before continuing.
The named feature is missing	Confirm the correct assigned account, English display language and app version. Use the stated manual practice route; record feature unavailable. This is not evidence that a native skill ran.

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Challenge

Repeat the task for one different site and explain how the filter scope changes the result. Keep the original verified output.

Reflection

Which evidence proves the output is correct, and which decision still needs a person?

Note: The lab folder labs/lab-01-*/ holds the instruction PDF with sample prompts, a data folder with templates, checkpoints and the mock-data workbook (data/Northstar_Work_Orders.xlsx). Northstar Water Services is a fictional water-pipe maintenance and service company. All service hubs, staff and work orders are synthetic. Use the fixed reporting date 2026-09-30; missing facts remain UNKNOWN. Proposals do not change real assignments. Training email drafts are never sent.

Concepts for Lab 2: AI Security: Five Principles, Prompts and Verified App Handoffs, Outlook Drafting and Operational Decisions.

AI Security: Five Principles

AI security is the set of habits and controls that keep data, decisions and systems safe when people work with AI assistants and agents. Five principles apply to every Copilot task in this course: data control, permission control, human in the loop, traceability, and using a well-tested model rather than the newest one for important work.

Copilot only surfaces data a user can already access and does not use prompts to train its foundation models, but people can still overshare, give an agent too much access, accept an unchecked draft or lose track of what was used. The principles turn those risks into checks you can perform before work leaves your hands.

How it works

- Share only the data the task needs.
- Limit access for people, agents and skills.
- Review and approve before anything takes effect.
- Log sources, prompts, versions and checks.

Worked example

- Use synthetic or sanitised data in training and demos.
- An agent gets Can chat access, not Can edit, unless needed.
- A Copilot email stays a draft until a person reviews it.

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Decision guide

Use when	Avoid when
The output could reach other people, systems or decisions.	Confidential data is pasted into an unapproved tool or a draft is sent unchecked.

Authoritative references

- https://learn.microsoft.com/en-us/microsoft-365/copilot/microsoft-365-copilot-privacy
- https://learn.microsoft.com/en-us/microsoft-365/copilot/microsoft-365-copilot-overview

Prompts and Verified App Handoffs

A useful prompt combines the intended goal, operational context, named source and output expectations. The handoff object is a verified artifact, such as a Word brief with a checked metric table and source links. It is not an assumption that another app can remember the preceding conversation. For maintenance, expectations should specify exact dates, status vocabulary, current versus proposed assignments and missing-value behaviour.

The same number may have different meaning when a chart counts all work orders while an email calls it open backlog. A cross-app evidence register records the claim, value, source table, filter scope and date. This provides a small reproducible contract for each output and lets a reviewer identify a mismatch before it becomes a management decision.

How it works

- Write the goal and audience.
- Attach or select the actual source.
- Specify definitions and missing-value behaviour.
- Reconcile the returned claims.

Worked example

- Goal: draft a maintenance update for the operations manager.
- Source: reviewed Northstar operations brief, all filters All, 30 September.
- Expectation: label assignment suggestions as proposals and leave unknown approvals UNKNOWN.

Decision guide

Use when	Avoid when
A reviewed brief can be selected or its checked text pasted.	Only an unverified summary is available.

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Authoritative references

- https://support.microsoft.com/en-gb/word/copilot/draft-and-add-content-with-copilot-in-word
- https://support.microsoft.com/en-us/powerpoint/copilot/add-a-slide-from-a-file-with-copilot-in-powerpoint
- https://support.microsoft.com/en-us/microsoft-365-copilot/file-formats-supported-by-microsoft-365-copilot

Outlook Drafting and Operational Decisions

Copilot can help structure a draft email from supplied context. A synthetic text thread is quoted context, not a live Outlook conversation; the course explicitly uses it that way. The writer remains responsible for recipients, claims, attachments, tone and the distinction between a proposed action and an approved instruction. Saving a draft does not send it or authorise a work-order change.

An apparently routine update can accidentally promise technician reassignment or imply that a manager approved a recovery plan. Current assignment is recorded data; proposed assignment is a separate planning output. Available hours are one input to planning, not evidence of approval or guaranteed productivity. The final email should request the decision rather than claim it has happened.

How it works

- Provide the synthetic thread and checked facts.
- Draft a concise update.
- Review recipients, facts and action wording.
- Save the unsent draft and evidence.

Worked example

- Current owner blank; proposed owner Aisha remains unapproved.
- A draft addressed to example.invalid is a training artifact.
- The handoff says decision requested, not decision approved.

Decision guide

Use when	Avoid when
You can check the evidence and review the draft before use.	The task would imply a real approval or send without review.

Authoritative references

- https://support.microsoft.com/en-us/microsoft-365-copilot/file-formats-supported-by-microsoft-365-copilot

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Lab 2 · Word to PowerPoint and Outlook — Create the Cross-App Maintenance Update

Learning outcome: LO1: integrate verified Word, Excel, PowerPoint and Outlook outputs.

Goal: Continue the verified operations brief from Lab 1. You will turn it into an initial PowerPoint update, then prepare an Outlook reply to a synthetic maintenance thread. You will test how grounding differs between an attached file, a chat transcript and a reviewed summary, and reconcile the same facts across all four apps.

Continue the verified operations brief from Lab 1. You will turn it into an initial PowerPoint update, then prepare an Outlook reply to a synthetic maintenance thread. You will test how grounding differs between an attached file, a chat transcript and a reviewed summary, and reconcile the same facts across all four apps.

What you'll build

Northstar_Initial_Update.pptx, an unsent Outlook draft and an Evidence_Log with Excel, Word, PowerPoint and Outlook figures reconciled. (Tools: Microsoft 365 Copilot · Excel · Word · PowerPoint · Outlook · OneDrive.)

Prerequisites

- Microsoft 365 account with Copilot available in Word, Excel, PowerPoint and Outlook. Availability depends on the assigned licence and tenant settings; confirm access with the trainer before class.
- Desktop Excel on Windows or Mac is recommended; save working copies to your assigned OneDrive course folder with AutoSave on. Do not overwrite the supplied input.
- Download this whole lab folder. All files are synthetic; work-genai is your own learner output folder.
- Verified Lab 1 workbook and Word brief, or supplied checkpoint and snapshot.

Step-by-step

1. Open your Lab 1 Word brief and cleaned workbook. If needed use this lab's verified checkpoint and snapshot, clearly recording rejoin. Read the synthetic email thread quoted in the Step 4 prompt. It asks for a portfolio update, not authorisation to reassign technicians.

```
WORD COPILOT:

From this selected verified Northstar Water Services operations brief,
extract a handoff table with reporting date, filter scope, total, closed,
cancelled, backlog, overdue, unassigned backlog and recorded cost. Separate
facts, proposed next actions and open decisions. Keep every value traceable
to the supplied brief.
```


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2. Open data/templates/Northstar_Management_Deck_Template.pptx in PowerPoint and save as Northstar_Initial_Update.pptx. For this initial update use four slides by keeping the first four layout slides and deleting the remaining unused template slides. Open Copilot and select your Word brief with Reference a file; if the command is unavailable, paste the reviewed brief into the prompt.

```
POWERPOINT COPILOT:

Using this verified Northstar Word brief, create a four-slide initial
update: scope and portfolio; backlog and overdue work; current assignments
and proposed actions; decision required. Use the existing white/navy/blue
design. Preserve all verified metrics. Add source/date notes. Do not invent
completed work, savings or approved reassignment.
```

3. Review all four slides. Check status definitions, units and date, and correct any invented claim. Record the reconciliation in the Evidence_Log sheet: one row per claim and a source link. This is the bridge between app outputs; app chats are not automatically shared.

```
POWERPOINT COPILOT:

Audit these four Northstar Water Services slides against the selected
verified operations brief. For each numeric claim give the source, date and
scope; flag unsupported service-restoration, safety, customer-impact or
approval claims. Propose corrections without changing the checked values.
```

4. In Outlook create a new training draft addressed to operations.manager@example.invalid with subject DRAFT - Northstar maintenance update - review required. Do not send. Paste the synthetic thread into the draft or use it as quoted context; Copilot summarises the supplied text, not a real mailbox thread. Open Copilot > Draft with Copilot; UI names may differ.

```
OUTLOOK COPILOT:

Draft a concise reply to the synthetic thread quoted below, using the
verified Northstar brief and these checked metrics. State the 30 September
2026 scope, backlog/overdue findings, proposed next actions and the
decision requested. Explain that assignments are unapproved proposals. Do
not claim the attachments have been sent. Keep as a draft.



THREAD (synthetic, do not send):

2026-09-29 09:00 Operations manager: Please prepare the 30 September
portfolio snapshot; show open work by site, category and current
assignment. 2026-09-29 10:15 Supervisor: Check overdue jobs before
proposing reassignment. Available hours are planning inputs, not approval.
2026-09-30 08:30 Analyst: The source export is in the workbook. No new
assignment has been approved. Please produce a Word brief and a short
PowerPoint update, then prepare a draft reply.
```


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5. If your Outlook offers reference attachments, attach the verified Word brief and choose it as context. Otherwise paste the reviewed metrics into the drafting prompt. Replace any placeholder summary with exact checked values; inspect recipients, subject, tone and attachment names. Save only the draft and a local TXT copy of its reviewed body.

6. Ask Word Copilot to compare your pasted slide/email summaries with the brief, then perform the same comparison yourself. Add the cross-app prompt log entries manually. Keep all four source links with the reconciliation record; any mismatch must be fixed before the skill labs.

```
WORD COPILOT:

Compare these pasted PowerPoint and Outlook summaries with this verified
Word brief. List inconsistent figures, status definitions, dates, owner
claims or approved-action wording. Do not resolve a conflict by guessing.
Return a discrepancy table with source and correction needed.
```

Sample prompts

Purpose	Prompt
Scope	State actual source, report date and selected scope before acting.
Verify	Explain the source/formula checks and preserve unknowns and original records.

Test it

Check	Expected result
Four apps	Excel Checks, Word brief, four-slide update and unsent Outlook draft share the same date and values.
Cross-app grounding	Named file or pasted verified evidence is provided in each app; no assumed shared chat.
Draft boundary	example.invalid recipient; nothing sent and no assignment approved.

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Checkpoint and rejoin point

If you fell behind, use data/checkpoints/Northstar_Verified_Checkpoint.xlsx and data/checkpoints/Northstar_Verified_Snapshot.docx. Record checkpoint used in Prompt_Log; then perform this lab, rather than claiming the checkpoint as your output.

Troubleshooting

If this happens	Fix
Copilot cannot find the source	Use the app file picker to select the actual OneDrive file. A typed local path or a different app chat is not shared context. If needed copy only the verified snapshot into the current app.
A total disagrees with Control	Clear dashboard filters to All, check table extent and duplicate IDs, recalculate, then compare definitions and reporting date. Reconcile before continuing.
The named feature is missing	Confirm the correct assigned account, English display language and app version. Use the stated manual practice route; record feature unavailable. This is not evidence that a native skill ran.

Challenge

Repeat the task for one different site and explain how the filter scope changes the result. Keep the original verified output.

Reflection

Which evidence proves the output is correct, and which decision still needs a person?

Note: The lab folder labs/lab-02-*/ holds the instruction PDF with sample prompts, a data folder with templates, checkpoints and the mock-data workbook (data/Northstar_Work_Orders.xlsx). Northstar Water Services is a fictional water-pipe maintenance and service company. All service hubs, staff and work orders are synthetic. Use the fixed reporting date 2026-09-30; missing facts remain UNKNOWN. Proposals do not change real assignments. Training email drafts are never sent.

Topic 1 Recap - Mapped to Learning Outcomes

Use this checkpoint to explain what you can now do and identify any lab evidence you still need to repair.

Learning outcome	Evidence from this topic
LO1	Same work-order evidence across four apps.
Verify	120 clean IDs and reconciled brief/update/draft.

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Topic 2 — Personalisation, .Rules and Skills for Excel

Account preferences, workbook conventions and reusable methods

Key concepts

- Personalisation — Your account preferences, applied across every workbook.
- .Rules — Visible column-A instructions that travel with the workbook.
- Skills — Reusable methods: built-in @chart-design, or your own SKILL.md such as Pareto and dashboard Skills.

Concepts for Lab 3: Account Personalisation and Specific Requests.

Account Personalisation and Specific Requests

Excel Copilot Personalisation stores account-level preferences for display, writing style and formula conventions. Saved preferences apply to subsequent prompts across workbooks and devices using that account. They are not stored in or shared with a workbook. A specific task instruction can supersede a preference for that task.

Consistent S$ display and short explanations reduce repetitive requests without changing numeric values. Preferences must not replace the workbook's business definitions. A private before-state note lets learners restore their previous account settings after the controlled exercise.

How it works

- Choose recurring display preferences.
- Save them to the account.
- Compare two workbook outputs.
- Test a specific request and restore.

Worked example

- A value of 35827.5 can display as S$35,827.50 or S$35,827.5000.
- Changing a number format does not change its numeric value.
- Another user opening the file does not receive your account preferences.

Decision guide

Use when	Avoid when
The preference applies to many reports on your account.	The instruction is a workbook-specific business rule or source change.

Authoritative references

- https://support.microsoft.com/en-us/excel/copilot/copilot-in-excel-personalization

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Lab 3 · Personalisation — Personalise Copilot in Excel for Consistent Service Reports

Learning outcome: LO2: distinguish account Personalisation, workbook .Rules and reusable Skills; configure, apply and verify each using water-service evidence..

Goal: Configure temporary account-level display and explanation preferences, test them in two copies of the same verified Northstar workbook, and distinguish preferences from workbook-specific definitions. A task-specific prompt can supersede Personalisation. Restore your previous settings after the exercise.

Configure temporary account-level display and explanation preferences, test them in two copies of the same verified Northstar workbook, and distinguish preferences from workbook-specific definitions. A task-specific prompt can supersede Personalisation. Restore your previous settings after the exercise.

What you'll build

Northstar_Personalisation_Test.xlsx, a two-workbook preference comparison and a private note confirming settings restored. (Tools: Microsoft 365 Copilot in Excel · OneDrive.)

Prerequisites

- Microsoft 365 account with Copilot available in Word, Excel, PowerPoint and Outlook. Availability depends on the assigned licence and tenant settings; confirm access with the trainer before class.
- Desktop Excel on Windows or Mac is recommended; save working copies to your assigned OneDrive course folder with AutoSave on. Do not overwrite the supplied input.
- Download this whole lab folder. All files are synthetic; work-genai is your own learner output folder.

Step-by-step

1. Open this lab's clean data/Northstar_Work_Orders.xlsx and save working copies as Northstar_Personalisation_Test.xlsx and Northstar_Preferences_Copy.xlsx in work-genai. Confirm 120 unique IDs, reporting date 2026-09-30 and cost S$35,827.50. Neither copy needs a .Rules sheet or an installed Skill for this test.

```
EXCEL - CHAT ONLY:

Inspect tblWorkOrders and Control for Northstar Water Services. State the
clean row count, reporting date and recorded cost. Explain why display
preferences should not change source values or reporting definitions. Do
not edit.
```

2. In Copilot select Settings (...) > Personalization. Keep your existing settings in a private local note, then temporarily enter these four preferences: Use S$ with two

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decimal places and thousands separators for currency displays. Display dates as DD-MMM-YYYY. Prefer structured references when writing Excel formulas. Start explanations with a short plain-English summary before the detail. Select Save. Preferences apply to your account on subsequent prompts; they are not stored in or shared with the workbook.

3. In the first working copy request a small output using your saved preferences. Review the plan before allowing editing. Confirm the date displays 30-Sep-2026 in both copies. Inspect the formulas and number formats in the new Preference_Check sheet; do not accept rounded or rewritten source values.

```
EXCEL - PLAN:

Plan a Preference_Check sheet showing Control!B2 reporting date, total work
orders and recorded Cost_SGD from tblWorkOrders for Northstar Water
Services. Use my saved display and explanation preferences. Preserve
Work_Orders, IDs, owners and exact values.
```

```
EXCEL - ALLOW EDITING:

Apply the reviewed Preference_Check plan with structured-reference
formulas. Show a short explanation first and leave the source unchanged.
```

4. Open the second copy using the same signed-in account and a new Copilot chat. Do not repeat the preference text in this request. Compare both output displays and formula references; record actual behaviour in the Evidence_Log sheet. A missed preference is a finding to clarify, not proof of a guaranteed rule.

```
EXCEL - ALLOW EDITING:

In this second Northstar workbook, create Preference_Check with formula-
backed Control!B2 reporting date, total work orders and recorded Cost_SGD.
Follow my saved preferences, preserve source records and explain the result
briefly.
```

5. Return to the first workbook. Test a specific task instruction that changes only the result display to four decimal places. Verify S$35,827.5000 displays while the numeric value remains 35827.5. Record the comparison; do not alter the saved account settings or source.

```
EXCEL - ALLOW EDITING:

For this task only, display the recorded cost result in Preference_Check
with four decimal places. Keep its formula and exact numeric value
unchanged. Do not modify source cells or my saved preferences.
```

6. Explain the scope distinction, log the prompts and your observed checks, then return to Settings (...) > Personalization, restore your previous preferences and Save. Keep the restoration note private; do not upload personal account settings to the shared course folder.

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```
EXCEL - CHAT ONLY:

Explain which choices in this exercise belong to my account, which belong
to this Northstar workbook and which apply only to one prompt. Use the
results I actually observed. Do not claim my preferences are shared with
another person or added to the workbook.
```

Sample prompts

Purpose	Prompt
Scope	State actual source, report date and selected scope before acting.
Verify	Explain the source/formula checks and preserve unknowns and original records.

Test it

Check	Expected result
Source controls	Both copies retain120 unique IDs, original owners/statuses and exact cost 35827.50.
Display check	Saved S$ two-decimal/date/explanation preferences are checked in two workbooks; observed misses are recorded.
Specific request	The output can show35,827.5000 without changing its value or the saved preferences.
Restoration	Previous account preferences restored; private baseline note stays outside public courseware.

Checkpoint and rejoin point

If you fell behind, use data/checkpoints/Northstar_Verified_Checkpoint.xlsx and data/checkpoints/Northstar_Verified_Snapshot.docx. Record checkpoint used in Prompt_Log; then perform this lab, rather than claiming the checkpoint as your output.

Troubleshooting

If this happens	Fix
Copilot cannot find the source	Use the app file picker to select the actual OneDrive file. A typed local path or a different app chat is not shared context. If needed copy only the verified snapshot into the current app.
A total disagrees with Control	Clear dashboard filters to All, check table extent and duplicate IDs, recalculate, then compare definitions and reporting date. Reconcile before continuing.
The named feature is missing	Confirm the correct assigned account, English display language and app version. Use the stated manual practice route; record feature unavailable. This is not evidence that a native skill ran.

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Challenge

Repeat the task for one different site and explain how the filter scope changes the result. Keep the original verified output.

Reflection

Which evidence proves the output is correct, and which decision still needs a person?

Note: The lab folder labs/lab-03-*/ holds the instruction PDF with sample prompts, a data folder with templates, checkpoints and the mock-data workbook (data/Northstar_Work_Orders.xlsx). Northstar Water Services is a fictional water-pipe maintenance and service company. All service hubs, staff and work orders are synthetic. Use the fixed reporting date 2026-09-30; missing facts remain UNKNOWN. Proposals do not change real assignments. Training email drafts are never sent.

Concepts for Lab 4: Visible Workbook .Rules and Formula Results.

Visible Workbook .Rules and Formula Results

A visible worksheet named exactly .Rules, including the dot, supplies workbook-specific Copilot instructions in column A. Use one idea per cell and English for fully supported behaviour. The sheet travels with a copied workbook. Rules provide guidance for Copilot; they do not enforce formulas, permissions or security controls.

The workbook can carry stable scope, status, cost-display and preservation instructions alongside the source. A formula cell may supply a reporting-date instruction: Copilot reads its recalculated result on the next request. That does not automatically recalculate or update every AI-created output.

How it works

- Name the visible sheet exactly .Rules.
- Place one instruction per column-A cell.
- Recalculate a formula-driven rule.
- Check the next output and copied workbook.

Worked example

- The date rule returns Reporting date: 2026-09-30 from Control!B2.
- Hidden .Rules is not used; keep the final sheet visible.
- No universal precedence among rules, preferences and Skills is assumed.

Decision guide

Use when	Avoid when
Conventions belong to this workbook and should travel with it.	You need a security boundary or are assuming hidden rules apply.

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Authoritative references

- https://support.microsoft.com/en-us/excel/copilot/copilot-in-excel-rules
- https://support.microsoft.com/en-us/excel/copilot/copilot-in-excel-personalization
- https://support.microsoft.com/en-us/excel/copilot/copilot-in-excel-skills

Lab 4 · Workbook .Rules — Create and Test Workbook .Rules for Water Service Reporting

Learning outcome: LO2: distinguish account Personalisation, workbook .Rules and reusable Skills; configure, apply and verify each using water-service evidence..

Goal: Turn Northstar's workbook conventions into a visible .Rules worksheet. Keep one rule per cell in column A, include a formula-driven reporting-date rule, test the output, and verify that the sheet travels with a copied workbook. A .Rules sheet is guidance for Copilot, not Excel formula enforcement or a security boundary.

Turn Northstar's workbook conventions into a visible .Rules worksheet. Keep one rule per cell in column A, include a formula-driven reporting-date rule, test the output, and verify that the sheet travels with a copied workbook. A .Rules sheet is guidance for Copilot, not Excel formula enforcement or a security boundary.

What you'll build

Northstar_Rules_Workbook.xlsx with visible .Rules column A, a checked Rules_Test output and a copied-workbook sharing check. (Tools: Microsoft 365 Copilot in Excel · OneDrive.)

Prerequisites

- Microsoft 365 account with Copilot available in Word, Excel, PowerPoint and Outlook. Availability depends on the assigned licence and tenant settings; confirm access with the trainer before class.
- Desktop Excel on Windows or Mac is recommended; save working copies to your assigned OneDrive course folder with AutoSave on. Do not overwrite the supplied input.
- Download this whole lab folder. All files are synthetic; work-genai is your own learner output folder.
- Lab 3 scope comparison; verified Lab1 clean source or supplied checkpoint.

Step-by-step

1. Open the clean workbook, save as Northstar_Rules_Workbook.xlsx and inspect Control. In Copilot use Add work content > Create workbook rules if offered;

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otherwise insert a sheet manually and rename it exactly .Rules, including the leading dot. Keep it visible. Rules are fully supported only in English.

```
EXCEL - CHAT ONLY:

Inspect this Northstar workbook. Identify the source table, fixed reporting
date, status vocabulary and current owners. List workbook-specific
conventions that belong in .Rules rather than account Personalisation. Do
not edit.
```

2. Create the .Rules lines with the one-shot Allow editing prompt below (or copy them from the prompts page): one heading or rule per column-A cell; no data in other columns is required for recognition. Check source preservation, backlog/overdue/rate definitions, S$ display, colours, scope and controlled replacements.

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EXCEL - ALLOW EDITING:

Add a visible worksheet named exactly .Rules (with the leading dot) as the last tab; if it exists, clear column A and reuse it. Fill column A only, one cell per line below, entering only the text after each "A#: " label. Do not change any other sheet.

A1: ## SOURCE AND SCOPE

A2: Preserve Work_Orders, tblWorkOrders, Technicians, current Assigned_To values and source IDs.

A3: Use the reporting date from Control!B2. State the selected Site, Job_Category, Status and Assigned_To scope beside every report.

A4: Backlog excludes Closed and Cancelled. Overdue backlog has Due_Date strictly before Control!B2.

A5: On-time closure rate uses Closed rows only, with Closed_Date <= Due_Date. Display N/A when no Closed rows remain.

A6: ## FORMATS AND VISUALS

A7: Display recorded Cost_SGD as S$ with two decimals; keep underlying values exact. Show dates as DD-MMM-YYYY.

A8: Use white backgrounds, navy titles, blue and teal accents, readable labels and named units; never use colour alone.

A9: ## DASHBOARD LAYOUT

A10: Dashboard sheet: title in A1; Site, Job_Category, Status and Assigned_To dropdowns in B2:B5 (All plus source values); the six measures in B7:B12; data-as-at date and selected scope in D2.

A11: Six KPI tiles in D4:I7, one column per tile: title in row 4, large value in row 5, short definition in row 6 and "Checks: PASS" in row 7.

A12: Charts in a grid below the tiles: work orders by status and by job category side by side, then a full-width stacked bar by site (overdue backlog, other backlog, Closed, Cancelled).

A13: Slicers for Site, Job_Category, Status and Assigned_To on the right of the tiles, connected to tblWorkOrders. KPI formulas use SUBTOTAL visibility so slicers and dropdowns both filter every tile and chart.

A14: Put chart summaries, chart-title formulas and KPI checks on Dashboard_Summaries. Never merge cells. Hide gridlines on Dashboard.

A15: ## DASHBOARD STYLE

A16: Tiles: white bold text on solid navy #1F4E79 (totals and rates), blue #2E75B6 (backlog, overdue) or teal #2A9D8F (Closed, cost).

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```
A17: Every chart title states the finding from a cell formula, e.g. "Changi
Workshop has the most overdue backlog (18 of 50)".

A18: Show data labels on bars; no gridlines; rates as % with one decimal.
Each chart total must equal B7.

A19: ## OUTPUT AND REVIEW

A20: Use structured table references. Preserve the visible .Rules sheet and
unrelated sheets, charts and Prompt_Log.

A21: Keep current owners separate from unapproved assignment proposals.
Missing facts and approvals remain UNKNOWN.

A22: Ask before replacing named output sheets. Reconcile every count, rate
and chart against source rows; do not duplicate outputs on rerun.
```

```
EXCEL - PLAN:

Review the visible .Rules column A for this Northstar workbook. Identify
ambiguous instructions or conflicts with the actual source and Control.
Propose concise corrections, keeping one idea per cell. Do not edit the
source or invent precedence between rules, preferences and Skills.
```

3. After reviewing any correction, update only .Rules. Add a formula in the next empty column-A cell: ="Reporting date: "&TEXT(Control!B2,"yyyy-mm-dd"). Recalculate and confirm it displays Reporting date:2026-09-30. Copilot reads the formula result on the next prompt; this does not rewrite dashboard values by itself.

```
EXCEL - ALLOW EDITING:

Apply only my reviewed .Rules corrections. Keep the sheet visible, one
instruction per column-A cell, and retain the reporting-date formula.
Preserve source data, Control and unrelated sheets.
```

4. Request a Rules_Test summary without restating the formatting rules. Check counts against Control, source references, displayed date/scope, S$ units and proposal wording. The known baseline is120 total,60 backlog,50 overdue and 30/48 on-time Closed.

```
EXCEL - ALLOW EDITING:

Create Rules_Test for Northstar Water Services from tblWorkOrders using the
visible workbook .Rules. Include total, backlog, overdue, on-time Closed
rate and recorded cost with formula/source evidence. Show date and All
scope. Keep source owners and values unchanged.
```

5. Save As Northstar_Rules_Shared_Copy.xlsx. Open the copy and confirm .Rules is still present and visible in column A; you have not exported your account preferences. If testing a hidden sheet, use only a sandbox copy, hide .Rules, record

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that hidden rules are not used, then unhide before continuing. Do not claim native rule use when the feature is unavailable.

```
EXCEL - CHAT ONLY:

In this copied Northstar workbook, identify the visible .Rules instructions
and reporting-date formula result. Explain which guidance travels with the
workbook and which account preferences do not. Verify Rules_Test against
Control without changing records.
```

6. Record the sheet name, visibility, column, formula result, actual output checks and copied-workbook check in the Evidence_Log sheet. Carry the rules-enabled workbook forward. If a prompt, Skill and rule disagree, ask for clarification and verify the output; no universal precedence is assumed.

```
EXCEL - CHAT ONLY:

Compare Personalisation, .Rules and a reusable Skill for Northstar
reporting. Place currency preferences, this workbook's backlog/date rules
and the repeatable Pareto workflow in the right scope. Identify conflicts
requiring clarification and the checks a person still must perform.
```

Sample prompts

Purpose	Prompt
Scope	State actual source, report date and selected scope before acting.
Verify	Explain the source/formula checks and preserve unknowns and original records.

Test it

Check	Expected result
Recognition	Exactly .Rules; visible worksheet; instructions in column A, one idea per cell.
Formula rule	Reporting-date formula recalculates and displays2026-09-30; no source values changed.
Output controls	120 total;60 backlog;50 overdue;30/48 on-time;S$35,827.50 at All scope.
Workbook copy	Visible .Rules travels with the copy; account preferences are not exported.

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Checkpoint and rejoin point

If you fell behind, use data/checkpoints/Northstar_Verified_Checkpoint.xlsx and data/checkpoints/Northstar_Verified_Snapshot.docx. Record checkpoint used in Prompt_Log; then perform this lab, rather than claiming the checkpoint as your output.

Troubleshooting

If this happens	Fix
Copilot cannot find the source	Use the app file picker to select the actual OneDrive file. A typed local path or a different app chat is not shared context. If needed copy only the verified snapshot into the current app.
A total disagrees with Control	Clear dashboard filters to All, check table extent and duplicate IDs, recalculate, then compare definitions and reporting date. Reconcile before continuing.
The named feature is missing	Confirm the correct assigned account, English display language and app version. Use the stated manual practice route; record feature unavailable. This is not evidence that a native skill ran.

Challenge

Repeat the task for one different site and explain how the filter scope changes the result. Keep the original verified output.

Reflection

Which evidence proves the output is correct, and which decision still needs a person?

Note: The lab folder labs/lab-04-*/ holds the instruction PDF with sample prompts, a data folder with templates, checkpoints and the mock-data workbook (data/Northstar_Work_Orders.xlsx). Northstar Water Services is a fictional water-pipe maintenance and service company. All service hubs, staff and work orders are synthetic. Use the fixed reporting date 2026-09-30; missing facts remain UNKNOWN. Proposals do not change real assignments. Training email drafts are never sent.

Concepts for Lab 5: Built-in and Custom Skills, Pareto Skill: Counts, Sorting and Cumulative Share.

Built-in and Custom Skills

Copilot in Excel ships with built-in skills you call with @, such as @chart-design for charts. @chart-design applies the Excel design team's chart guidance and @theme-design handles workbook formatting. Built-in skills are turned on in Settings (...) > Manage plugins & skills; chart-design sits under Visualization and is off by default. A

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custom skill is your own SKILL.md for a method Excel does not know, such as the water-service Pareto or dashboard.

Start with a built-in skill when the task is generic, such as making a clear chart. Write a custom skill when the method carries your own rules: which rows count, how ties sort, which outputs may be replaced and which checks must pass. Both are reusable; only the custom one knows Northstar.

How it works

- Turn on chart-design under Visualization.
- Call @chart-design for a generic chart.
- Spot the rules it cannot know.
- Write a custom SKILL.md for those rules.

Worked example

- @chart-design draws a clean overdue-by-category bar chart.
- It does not know the overdue population or Pareto tie order.
- water-service-pareto adds the cumulative line and named output.

Decision guide

Use when	Avoid when
A generic design task or your own repeatable method.	You expect a built-in skill to know your business rules.

Authoritative references

- https://support.microsoft.com/en-us/excel/copilot/copilot-in-excel-built-in-skills
- https://support.microsoft.com/en-us/excel/copilot/copilot-in-excel-skills

Pareto Skill: Counts, Sorting and Cumulative Share

A Pareto contribution chart orders category counts from largest to smallest and adds a cumulative-percentage line. The population here is overdue backlog, not all work orders: nonterminal statuses with Due_Date strictly before the reporting date. Alphabetical tie sorting keeps equal counts in a repeatable order.

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The Skill turns a recurring grouping-and-charting method into reusable instructions. Counts and cumulative formulas must reconcile to the same eligible source rows. A right-hand percentage axis from zero to 100% prevents interpreting the line as another count series. Category concentration describes contribution rather than establishing root cause.

How it works

- Define source, population, date and grouping.
- Count and sort with stable ties.
- Calculate cumulative count / eligible total.
- Build and verify an editable dual-axis chart.

Worked example

- Pump 18; Meter 12; Leak 8; Valve 8; Pipe 4: total 50.
- Cumulative percentages are 36%, 60%, 76%, 92%, 100%.
- Four of five categories first reach 80%; do not force an 80/20 conclusion.

Decision guide

Use when	Avoid when
The count population and grouping are defined and auditable.	Categories are being called causes or zero-total shares treated as measured 0%.

Authoritative references

- https://support.microsoft.com/en-us/excel/copilot/copilot-in-excel-skills

Lab 5 · Custom Pareto Chart Skill — Use the Built-in Chart Skill, then Create a Custom Pareto Skill

Learning outcome: LO2: distinguish account Personalisation, workbook .Rules and reusable Skills; configure, apply and verify each using water-service evidence..

Goal: Author water-service-pareto/SKILL.md with Copilot, install it in Excel and apply it to the same overdue work-order queue. Build a descending count chart plus cumulative-percentage line, then check a controlled rerun and a second grouping. Contribution analysis does not establish root cause or force an 80/20 split.

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Start with Excel's built-in @chart-design skill, then author water-service-pareto/SKILL.md with Copilot, install it in Excel and apply it to the same overdue work-order queue. Build a descending count chart plus cumulative-percentage line, then check a controlled rerun and a second grouping. Contribution analysis does not establish root cause or force an 80/20 split.

What you'll build

water-service-pareto/SKILL.md, Northstar_Pareto.xlsx with Pareto_Job_Category and Pareto_Site, and a verified rerun record. (Tools: Excel Copilot · Microsoft 365 Copilot web · OneDrive.)

Prerequisites

- Microsoft 365 account with Copilot available in Word, Excel, PowerPoint and Outlook. Availability depends on the assigned licence and tenant settings; confirm access with the trainer before class.
- Desktop Excel on Windows or Mac is recommended; save working copies to your assigned OneDrive course folder with AutoSave on. Do not overwrite the supplied input.
- Download this whole lab folder. All files are synthetic; work-genai is your own learner output folder.
- Lab 4 visible .Rules and clean source; assigned Excel Skill feature checked.

Step-by-step

1. Open your rules-enabled workbook from Lab 4 or this lab's clean rules-enabled data/Northstar_Work_Orders.xlsx. Save as Northstar_Pareto.xlsx. Read the expected Pareto controls in Test It and the supplied Skill template before drafting your own file. Work only with synthetic data.

```
EXCEL - CHAT ONLY:

For tblWorkOrders at Control!B2, All sites, count overdue backlog by
Job_Category: Status neither Closed nor Cancelled and Due_Date strictly
before the reporting date. Show category counts and total before charting.
Preserve all records.
```

2. Built-in skill first. In Excel Copilot open Settings (...) > Manage plugins & skills. Under Visualization turn on chart-design (it is off by default) and confirm it is enabled. Ask @chart-design for a quick chart of the same overdue counts. Note what the built-in skill gives you (chart design guidance) and what it does not know (your overdue population, tie order, cumulative line and named output) - that is why you build the custom water-service-pareto Skill next. Keep the demo sheet or undo it.

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```
EXCEL - ALLOW EDITING:

@chart-design Chart the overdue backlog by Job_Category from tblWorkOrders
at Control!B2 as a bar chart, longest bar first, on a new sheet named
ChartDesign_Demo. Use a finding-style title, axis titles and data labels.
Do not change source data.
```

3. In Microsoft 365 Copilot on the web at https://copilot.cloud.microsoft, provide this self-contained specification. Copy the complete response into a plain-text editor and save UTF-8 SKILL.md inside a matching water-service-pareto folder. Compare it to data/templates/water-service-pareto/SKILL.md. If short of time, use the reviewed supplied file and record that choice.

```
MICROSOFT 365 COPILOT WEB:

Write a complete SKILL.md named water-service-pareto for a recurring water-
pipe service report. Require YAML name and description, actual source
table, reporting date, scope, count metric and grouping column. Source
tblWorkOrders has Work_Order_ID, Job_Category, Site, Status and Due_Date.
Overdue backlog excludes Closed/Cancelled and has Due_Date before the
supplied date. Read visible workbook .Rules; preserve source
IDs/statuses/owners and unrelated sheets. Count by grouping, sort count
descending then label alphabetically for ties, and create formula-backed
count, cumulative count, share and cumulative percentage. Build an editable
combo chart: count columns and cumulative line on a 0%-100% right axis.
Output only Pareto_<grouping column>; ask before replacing the matching
output, rerun without duplicates, and leave zero-total percentages N/A.
Verify counts, stable ties and final 100%; do not infer causes or force
80/20. Return the entire file.
```

4. In Excel use Copilot Settings (...) > Manage plugins & skills > Custom skills > Create OneDrive folder if needed > Open skills folder. Copy water-service-pareto/SKILL.md into that app-created location, Refresh and enable it. Verify the name using Choose plugins & skills or @mention. If unavailable, paste the complete reviewed Skill before each request, remove @water-service-pareto and record Manual practice - native Skill unavailable. Log web authoring manually.

5. Request a plan before building. Confirm the eligible total 50 and the five ordered category counts. After review, apply the Skill. Inspect Pareto_Job_Category, formulas, axes and source preservation; copying a static image does not meet the editable-chart goal.

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```
EXCEL - PLAN:

@water-service-pareto Plan an overdue-backlog Pareto Chart by Job_Category
from tblWorkOrders at 2026-09-30, All sites. Follow visible .Rules. Explain
summary formulas, descending counts, alphabetical ties, cumulative
percentages and the named output to create. Do not edit yet.


EXCEL - ALLOW EDITING:

@water-service-pareto Apply the reviewed Job_Category Pareto plan. Create
Pareto_Job_Category with a formula-backed summary and editable count-
column/cumulative-percent-line combo chart. Keep source data and unrelated
sheets unchanged.
```

6. Check counts against the supplied controls. Verify cumulative percentages 36%,60%,76%,92%,100% and two equal8-count categories in alphabetical order. The first four categories reach at least80%; do not call this proof that20% of categories cause80% of failures. If combo chart UI is absent, select the summary and use Insert > Combo > Custom Combination: Count clustered column, Cumulative_Percent line on secondary axis, right bounds 0 and 1; label both axes. Record a missing feature honestly.

```
EXCEL - CHAT ONLY:

Verify Pareto_Job_Category against eligible source rows. Check total 50,
descending counts, alphabetical tie order, cumulative percentages
36/60/76/92/100%, final 100% and labelled dual axes. State which category
set first reaches 80%, and why this is contribution rather than a causal
diagnosis.
```

7. Save a sandbox copy, rerun the same grouping after approving replacement of only Pareto_Job_Category, and compare source IDs/statuses/owners and sheet/chart counts. Then apply the Skill to Site to produce a distinct Pareto_Site output. Keep both reviewed summaries for later decision evidence.

```
EXCEL - PLAN:

@water-service-pareto Refresh the existing Pareto_Job_Category for the same
source, date and All scope. Identify exactly what will be replaced and ask
before replacement. Preserve every unrelated sheet and do not duplicate
charts.


EXCEL - ALLOW EDITING:

@water-service-pareto After the approved category rerun, create a separate
overdue-backlog Pareto_Site for the same source/date/scope. Preserve
Pareto_Job_Category and reconcile the new summary to 50 overdue orders.
```


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Sample prompts

Purpose	Prompt
Scope	State actual source, report date and selected scope before acting.
Verify	Explain the source/formula checks and preserve unknowns and original records.

Test it

Check	Expected result
Built-in skill	chart-design enabled under Visualization; @chart-design demo chart created without changing the source.
Eligible population	Overdue backlog50; no Closed or Cancelled records included.
Category counts	Pump Service=18; Meter Service=12; Leak Repair=8; Valve Repair=8; Pipe Replacement=4
Cumulative percentages	36%,60%,76%,92%,100%; Leak Repair precedes Valve Repair in the8-count tie.
Chart and rerun	Editable columns plus cumulative line; right axis0%-100%; matching output replaced only after review, without duplicates.
Source and interpretation	120 original records preserved; Site grouping also totals50; no invented root cause.

Checkpoint and rejoin point

If you fell behind, use data/checkpoints/Northstar_Verified_Checkpoint.xlsx and data/checkpoints/Northstar_Verified_Snapshot.docx. Record checkpoint used in Prompt_Log; then perform this lab, rather than claiming the checkpoint as your output.

Troubleshooting

If this happens	Fix
Copilot cannot find the source	Use the app file picker to select the actual OneDrive file. A typed local path or a different app chat is not shared context. If needed copy only the verified snapshot into the current app.
A total disagrees with Control	Clear dashboard filters to All, check table extent and duplicate IDs, recalculate, then compare definitions and reporting date. Reconcile before continuing.
The named feature is missing	Confirm the correct assigned account, English display language and app version. Use the stated manual practice route; record feature unavailable. This is not evidence that a native skill ran.

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Challenge

Repeat the task for one different site and explain how the filter scope changes the result. Keep the original verified output.

Reflection

Which evidence proves the output is correct, and which decision still needs a person?

Note: The lab folder labs/lab-05-*/ holds the instruction PDF with sample prompts, a data folder with templates, checkpoints and the mock-data workbook (data/Northstar_Work_Orders.xlsx), plus data/templates/water-service-pareto/SKILL.md (Reviewed Pareto Skill fallback). Northstar Water Services is a fictional water-pipe maintenance and service company. All service hubs, staff and work orders are synthetic. Use the fixed reporting date 2026-09-30; missing facts remain UNKNOWN. Proposals do not change real assignments. Training email drafts are never sent.

Concepts for Lab 6: Skills, Prompts and Workbook Rules, Custom Skill Discovery in Each App, Repeatability and Safe Skill Reruns, Assignment Proposals and Capacity Evidence.

Skills, Prompts and Workbook Rules

A prompt is one request. A custom skill is a reusable set of instructions that tells Copilot when and how to perform a recurring task. SKILL.md contains required YAML name and description followed by the instructions. Each skill needs its own folder with a matching name. Workbook-specific rules describe constraints for one workbook; a skill can carry the same maintenance process across repeated reports and supported apps.

The skill makes stable definitions and review boundaries reusable: backlog excludes two terminal statuses, overdue uses a fixed date, missing owners stay missing and output reruns do not duplicate sheets. It does not supply absent work-order data or turn an unsupported app feature into an available one. Skill availability must be verified separately in the app where it will run.

How it works

- Identify a recurring task.
- Separate stable rules from current inputs.
- Write name, description and instructions.
- Install, invoke and inspect the result.

Worked example

- maintenance-reporting contains definitions and output boundaries.
- Each request still supplies this month’s source, date and scope.
- Native skill use and pasted manual instructions are recorded differently.

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Decision guide

Use when	Avoid when
The same reporting method recurs and can be tested.	Inputs or desired behaviour are still undefined.

Authoritative references

- https://support.microsoft.com/en-us/excel/copilot/copilot-in-excel-skills
- https://support.microsoft.com/en-us/powerpoint/copilot/copilot-in-powerpoint-skills

Custom Skill Discovery in Each App

Excel uses its Manage plugins & skills interface and an app-created OneDrive skills folder. PowerPoint uses Choose skills and Manage skills with an Add skill/upload route or its own custom skills folder. The app UI resolves the supported folder; a guessed universal path is unreliable. Refresh and enable the skill, then verify the exact name in each app before invoking it by @mention or the menu.

Folder-name/frontmatter mismatches, an extra .txt extension, unavailable licence features and disabled skills can all prevent discovery. A web Copilot authoring conversation can produce the complete Markdown text, but the learner still saves, installs and tests it. Excel cannot log the web chat automatically. Clear installation evidence prevents confusing a manually pasted policy with a native reusable skill.

How it works

- Save complete UTF-8 SKILL.md.
- Use the app-created skills location.
- Refresh and verify the matching name.
- Run and record a smoke test.

Worked example

- Folder maintenance-reporting matches YAML name maintenance-reporting.
- Excel and PowerPoint both show the skill before native invocation.
- If a menu is absent, manual practice is labelled feature unavailable.

Decision guide

Use when	Avoid when
The app exposes its supported skill installation controls.	A missing menu is being treated as successful installation.

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Authoritative references

- https://support.microsoft.com/en-us/excel/copilot/copilot-in-excel-skills
- https://support.microsoft.com/en-us/powerpoint/copilot/copilot-in-powerpoint-skills

Repeatability and Safe Skill Reruns

Repeatability means the same verified inputs and method yield equivalent business measures and a controlled output structure. A safe rerun preserves source IDs, statuses, assignments and unrelated sheets. It creates or replaces only explicitly named outputs after review. The concept is similar to idempotence: rerunning should not append duplicate summaries, inflate totals or silently destroy a learner's unrelated work.

Reporting skills may be used every week or month, so uncontrolled output accumulation quickly creates conflicting summaries. An input guard asks for missing data; a replacement guard identifies the output to change. Test those guards using a copy and compare row count, sums, sheet inventory and output measures before and after, rather than relying on Copilot saying it was careful.

How it works

- Capture the source fingerprint and outputs.
- Request a reviewed rerun plan.
- Replace only named output areas.
- Compare source and measures again.

Worked example

- 120 source IDs remain 120 after rerunning Maintenance_Summary.
- Missing report date causes a clarification request, not invented numbers.
- A second output sheet is a defect when replacement was requested.

Decision guide

Use when	Avoid when
Named output boundaries and reconciliation checks exist.	The skill can overwrite source rows or unrelated sheets.

Authoritative references

- https://support.microsoft.com/en-us/excel/copilot/copilot-in-excel-skills
- https://support.microsoft.com/en-us/powerpoint/copilot/copilot-in-powerpoint-skills

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Assignment Proposals and Capacity Evidence

Assignment planning combines the work category, existing owner, estimated hours, technician primary category and available hours. These inputs help generate a proposal; they do not prove complete operational suitability, approved schedules or safety authorisation. Keep Current_Assigned_To separate from Proposed_Assigned_To and use an explicit manager-decision field. Missing evidence is UNKNOWN, not a confident allocation.

The same skill can improve Excel analysis and PowerPoint wording by preserving this boundary. Excel shows category/capacity checks and assumptions in a proposals sheet; PowerPoint explains the findings and requests a decision. A polished visual must not conceal uncertain capacity or make the proposed owner look like the current owner.

How it works

- Identify unassigned backlog.
- Compare category and available hours.
- Record the proposal and assumptions.
- Request the manager decision.

Worked example

- Unassigned work is counted only within nonterminal statuses.
- Available hours minus proposed effort is a planning check.
- The source Assigned_To stays unchanged until a real authorised operational update.

Decision guide

Use when	Avoid when
A manager can review the proposal and missing constraints.	The report claims automatic approval or verified safety qualification.

Authoritative references

- https://support.microsoft.com/en-us/excel/copilot/copilot-in-excel-skills
- https://support.microsoft.com/en-us/powerpoint/copilot/copilot-in-powerpoint-skills

Lab 6 · Reporting Skill — Create, Use and Improve the Maintenance Reporting Skill

Learning outcome: LO2: distinguish account Personalisation, workbook .Rules and reusable Skills; configure, apply and verify each using water-service evidence..

Goal: Reuse the Skill-authoring and installation method from Lab 5 to create maintenance-reporting. Apply it in Excel and PowerPoint, test missing inputs and a

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controlled rerun, then refine v2 using actual evidence. The supplied reviewed template supports rejoining; it is not proof of native execution.

Reuse the Skill-authoring and installation method from Lab 5 to create maintenance-reporting. Apply it in Excel and PowerPoint, test missing inputs and a controlled rerun, then refine v2 using actual evidence. The supplied reviewed template supports rejoining; it is not proof of native execution.

What you'll build

Northstar_Skill_Report.xlsx with Maintenance_Summary and Assignment_Proposals, Northstar_Skill_Update.pptx, SKILL.md v2 and a before/after improvement log. (Tools: Excel and PowerPoint Copilot · Microsoft 365 Copilot web · OneDrive.)

Prerequisites

- Labs 1–2 verified brief and four-slide update; Lab 5 Skill-authoring method.

Step-by-step

1. Use Microsoft 365 Copilot on the web to draft maintenance-reporting/SKILL.md from the complete specification below. Review it against data/templates/maintenance-reporting/SKILL.md, save the complete file as v1 in work-genai, then install it using the same Excel app-created custom Skills folder and Refresh route practised in Lab5. If a native Skill menu is missing, use the full reviewed file as pasted instructions, remove @mention and label the route manual.

```
MICROSOFT 365 COPILOT WEB:

Write complete maintenance-reporting SKILL.md with YAML name/description.
Require actual tblWorkOrders or a verified Word snapshot, report date and
filter scope. Respect visible .Rules. Define backlog, overdue and Closed-
only on-time rate. Preserve IDs/statuses/current owners; keep assignment
proposals separate and approvals UNKNOWN. Create a verified Excel summary
and improve a supplied four- or eight-slide PowerPoint report from selected
evidence. Ask for missing inputs and before replacing only named outputs;
rerun without duplicates or source changes. Keep the existing design and
source/date notes. Never send email or approve assignments. Return the
entire file.
```

2. Copy the Lab 6 reviewed SKILL.md to your output folder and retain v1. Open this lab's clean workbook or your Lab 1 cleaned workbook. Save as Northstar_Skill_Report.xlsx. Read Technicians: primary category and available hours are synthetic planning inputs, not approvals. Check that your skill is enabled in Excel.

3. Invoke the skill in Plan or Chat only, review its proposed output boundaries, then allow editing. Inspect the resulting formulas and charts. Summary counts must reconcile to Control, and the source table must remain 120 unique records.

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```
EXCEL - PLAN:

@maintenance-reporting Build Maintenance_Summary for tblWorkOrders at 2026-
09-30 with all filters All. Include total, closed, cancelled, backlog,
overdue, on-time closed rate and recorded cost; show formulas and source
scope. Plan first. Preserve source records.


EXCEL - ALLOW EDITING:

Apply that reviewed summary plan. Add Assignment_Proposals for unassigned
backlog only. Use tblTechnicians primary category and available hours as
planning inputs; show category/capacity checks, current owner and proposed
owner separately. Leave manager decision UNKNOWN. Do not change
Assigned_To.
```

4. Check the unassigned backlog count against Control. A technician with the same category may still lack usable capacity: use available hours minus known proposed hours, flag conflicts, and do not convert a suggestion into confirmed assignment. Copy your verified summary into a copy of data/checkpoints/Northstar_Verified_Snapshot.docx, preserving measures and date.

```
EXCEL - CHAT ONLY:

@maintenance-reporting Review Assignment_Proposals against tblWorkOrders
and tblTechnicians. Check each suggested owner has the matching water-
service job category, capacity limitations are visible and original
Assigned_To values are unchanged. Leave manager approval UNKNOWN; do not
assign or schedule a technician.
```

5. In PowerPoint select the reviewed Word snapshot and open the four-slide update from Lab2 or data/checkpoints/Northstar_Initial_Update_Checkpoint.pptx. Use Copilot > + > Choose skills > Manage skills > Add skill to upload the reviewed maintenance-reporting file, or use the app's own supported Skills folder. Verify it in PowerPoint separately from Excel. If missing, paste the complete reviewed Skill plus snapshot as manual instructions and remove @mention. Apply the following refinement after review.

```
POWERPOINT COPILOT:

@maintenance-reporting Improve this existing four-slide Northstar update
from the selected verified Word snapshot. Preserve slide count, all values
and source/date/scope notes; use clear claim headlines and the existing
design. Keep assignment suggestions explicitly proposed and do not assume
access to Excel chat.
```

6. Create a new EMPTY workbook with no table, open a fresh Copilot chat and run a missing-input test for a different reporting period with no source selected. A proper skill asks for source/date/scope rather than inventing metrics. If it guesses, record the failure and revise the skill's input guard. Return to Northstar_Skill_Report.xlsx

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and Save As Northstar_Skill_Rerun_Test.xlsx. Run the second summary request in this copy of the report workbook, not in the empty workbook; only named outputs may be replaced after review, with no duplicated source or output sheets. Compare a before/after copy of Work_Orders sorted by Work_Order_ID for status, owner, dates and cost; every original source value must match.

```
EXCEL - CHAT ONLY:

@maintenance-reporting Prepare my next maintenance report. I need a
DIFFERENT reporting period from the course example and have not supplied
any source, date or scope. This empty workbook contains no work orders.
Tell me what you need; do not invent data.



EXCEL - PLAN:

@maintenance-reporting Rebuild the existing Maintenance_Summary for the
same table, date and scope. Describe what will be replaced, ask before
replacement and keep source/unrelated sheets. Do not duplicate outputs.
```

7. In a plain text editor improve SKILL.md: require an input guard, category/capacity evidence, source provenance per claim and a clear rerun rule. Retain YAML name maintenance-reporting and add a body version note v2. Re-upload/Refresh in BOTH Excel and PowerPoint. Record actual before/after changes, or already present - retained for safeguards that your v1 already contains. Do not fabricate differences. Rerun the smoke tests and compare values. If using the manual route, replace the pasted instructions in both apps.

```
MICROSOFT 365 COPILOT WEB:

Improve this supplied maintenance-reporting skill to v2 using only my
actual smoke-test and rerun findings pasted here. Add explicit missing-
input and no-duplicate-output guards. Keep Excel source and current owners
protected. Return the complete revised SKILL.md and a short before/after
change log; do not fabricate test evidence.
```

Sample prompts

Purpose	Prompt
Scope	State actual source, report date and selected scope before acting.
Verify	Explain the source/formula checks and preserve unknowns and original records.

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Test it

Check	Expected result
Source preserved	120 unique orders; current owner/status unchanged.
Skill outputs	Backlog 60; overdue 50; on-time closed 30/48; cost S$35,827.50.
Improvement evidence	Input guard and rerun tested; v1/v2 retained; PowerPoint four-slide output accurate.

Checkpoint and rejoin point

If you fell behind, use data/checkpoints/Northstar_Verified_Checkpoint.xlsx and data/checkpoints/Northstar_Verified_Snapshot.docx. Record checkpoint used in Prompt_Log; then perform this lab, rather than claiming the checkpoint as your output.

Troubleshooting

If this happens	Fix
Copilot cannot find the source	Use the app file picker to select the actual OneDrive file. A typed local path or a different app chat is not shared context. If needed copy only the verified snapshot into the current app.
A total disagrees with Control	Clear dashboard filters to All, check table extent and duplicate IDs, recalculate, then compare definitions and reporting date. Reconcile before continuing.
The named feature is missing	Confirm the correct assigned account, English display language and app version. Use the stated manual practice route; record feature unavailable. This is not evidence that a native skill ran.

Challenge

Repeat the task for one different site and explain how the filter scope changes the result. Keep the original verified output.

Reflection

Which evidence proves the output is correct, and which decision still needs a person?

Note: The lab folder labs/lab-06-*/ holds the instruction PDF with sample prompts, a data folder with templates, checkpoints and the mock-data workbook (data/Northstar_Work_Orders.xlsx), plus data/templates/maintenance-reporting/SKILL.md (Reviewed reporting Skill fallback). Northstar Water Services is a fictional water-pipe maintenance and service company. All service hubs, staff and work orders are synthetic. Use the fixed reporting date 2026-09-30; missing facts remain UNKNOWN. Proposals do not change real assignments. Training email drafts are never sent.

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Topic 2 Recap - Mapped to Learning Outcomes

Use this checkpoint to explain what you can now do and identify any lab evidence you still need to repair.

Learning outcome	Evidence from this topic
LO2	Personalisation, .Rules, built-in @chart-design, Pareto and reporting Skills tested.
Verify	Preserved source plus missing-input and rerun tests.

Topic 3 — Interactive Dashboards with Custom Skills

Create the Skill, build the dashboard and prove interactions

Key concepts

- Measures — Six formula KPIs with clear denominators and N/A rules.
- Visual design — KPI tiles and charts whose titles state the finding.
- Interactions — Dropdowns and slicers filter every tile and chart.
- Refresh — Known deltas in an isolated test copy.

Concepts for Lab 7: Custom Dashboard Skill as a Reusable Contract, Dashboard Measures and Denominators, Visual Dashboard Design for Maintenance.

Custom Dashboard Skill as a Reusable Contract

water-service-dashboard is a separate custom Skill whose instructions define inputs, six KPI formulas, four shared filter selectors, three editable charts and controlled output boundaries. Visible .Rules carries workbook conventions. The request still supplies the actual source, fixed date and chosen scope. Installing another reporting Skill does not install this dashboard method.

A reusable contract makes every KPI and chart obey the same AND filter logic and allows a safe refresh using table references. Input and replacement guards protect the source and existing Pareto evidence. Rerun evidence includes sheet/chart inventory and source checks, not just a claim that the Skill ran successfully.

How it works

- Author and review the complete SKILL.md.
- Install and verify the dashboard name.
- Invoke a plan and build the reviewed output.
- Test common filters, append refresh and reruns.

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Worked example

- Dashboard B2:B5 select site, category, status and owner.
- B7:B12 and all chart summaries use the identical selected rows.
- A 122-row sandbox is separate from the untouched 120-row baseline.

Decision guide

Use when	Avoid when
A recurring dashboard needs consistent measures and review guards.	Tenant Skill access or source preservation is being assumed.

Authoritative references

- https://support.microsoft.com/en-us/excel/copilot/copilot-in-excel-skills
- https://support.microsoft.com/en-us/excel/create-and-share-a-dashboard-with-excel-and-microsoft-groups

Dashboard Measures and Denominators

A dashboard measure needs a business definition, numerator, denominator where relevant, date rule and filter scope. Total orders describes the selected portfolio. Backlog is the selected nonterminal work. Overdue is a subset of backlog. On-time closure rate describes Closed rows only: closed on or before their due date divided by Closed rows. Recorded cost sums selected Cost_SGD; it is not a budget variance or inferred savings.

Changing the denominator can make an attractive rate meaningless. If the selected data contains no Closed rows, the on-time rate is N/A because it is undefined; a displayed 0% would falsely suggest failed delivery. A Measures dictionary and independent control sheet make the definitions inspectable. A fixed reporting date gives reproducible comparisons, while TODAY would change the snapshot over time.

How it works

- Define source fields and scope.
- Compute count and denominator.
- Handle empty selections explicitly.
- Reconcile against controls.

Worked example

- Closed-only rate is different from closed/total completion share.
- A due-today backlog item is not overdue under a strict before-date rule.
- All selected status/chart counts reconcile to total selected work orders.

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Decision guide

Use when	Avoid when
Measures have agreed definitions and evidence.	A zero denominator is being represented as a measured 0%.

Authoritative references

- https://support.microsoft.com/en-us/excel/create-and-share-a-dashboard-with-excel-and-microsoft-groups

Visual Dashboard Design for Maintenance

A dashboard should make an operational decision easier. KPI cards summarise magnitude; bar charts compare sites, categories or status counts. Labels, units and source/date context define what each visual means. Site backlog and total orders by site answer different questions and must have different names. Use a line chart for a genuine ordered time series, not unordered site names.

Consistent chart scales, short labels, whitespace and limited accents help a manager compare the work queue quickly. Colour alone is insufficient to distinguish statuses. A category count is evidence of frequency, not a causal diagnosis; a busy site is not necessarily performing badly. Choose the visual based on the question and include the limitation that prevents overinterpretation.

How it works

- Name the maintenance decision.
- Choose the measure and comparison.
- Use an appropriate labelled chart.
- Check meaning and readability.

Worked example

- Status count chart shows the portfolio mix.
- Site backlog chart excludes Closed and Cancelled.
- A category count does not establish the root cause of overdue jobs.

Decision guide

Use when	Avoid when
The visual has a clear audience and question.	Decorative charts hide an ambiguous metric.

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Authoritative references

- https://support.microsoft.com/en-us/excel/create-and-share-a-dashboard-with-excel-and-microsoft-groups

Lab 7 · Custom Dashboard Skill — Create and Apply a Custom Interactive Dashboard Skill

Learning outcome: LO3: build and validate an interactive Excel maintenance dashboard.

Goal: Create water-service-dashboard/SKILL.md for the Northstar maintenance workflow, install it in Excel, then invoke it to plan and build an interactive dashboard. Six KPIs and three charts share four filter selections. Workbook .Rules supplies scope, display and review constraints; the Skill supplies the repeatable method.

Create water-service-dashboard/SKILL.md for the Northstar maintenance workflow, install it in Excel, then invoke it to plan and build an interactive dashboard. Six KPIs and three charts share four filter selections. Workbook .Rules supplies scope, display and review constraints; the Skill supplies the repeatable method.

What you'll build

Northstar_Dashboard.xlsx with a Dashboard layout, six verified measures, three labelled charts and a measure dictionary ready for Lab 8 filters. (Tools: Excel Copilot · Microsoft 365 Copilot web · OneDrive.)

Prerequisites

- Microsoft 365 account with Copilot available in Word, Excel, PowerPoint and Outlook. Availability depends on the assigned licence and tenant settings; confirm access with the trainer before class.
- Desktop Excel on Windows or Mac is recommended; save working copies to your assigned OneDrive course folder with AutoSave on. Do not overwrite the supplied input.
- Download this whole lab folder. All files are synthetic; work-genai is your own learner output folder.
- Verified clean table from earlier labs or supplied clean checkpoint.

Step-by-step

1. Open the synthetic source and data/templates/water-service-dashboard/SKILL.md. In Microsoft 365 Copilot on the web draft the full reusable Skill from this specification. Save it inside a matching water-service-dashboard folder and review every guard/measure before installation. Log the web prompt manually.

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MICROSOFT 365 COPILOT WEB:

```
Write complete water-service-dashboard SKILL.md with YAML name/description.
Require actual tblWorkOrders and report date in Control!B2; read
visible .Rules. Preserve source IDs/statuses/owners, Technicians,
Control, .Rules, Prompt_Log, Pareto_* and unrelated sheets. Plan first and
ask before replacing only named dashboard outputs; rerun without duplicated
charts. Dashboard B2:B5 select Site, Job_Category, Status, Assigned_To with
All bypasses and AND logic. B7 total, B8 backlog excluding
Closed/Cancelled, B9 backlog Due_Date before Control!B2, B10 Closed, B11
on-time Closed_Date<=Due_Date among Closed only with N/A if none, B12
recorded Cost_SGD across selected statuses. Follow visible .Rules DASHBOARD
LAYOUT and STYLE: title A1; six KPI tiles in D4:I7 with Checks: PASS; three
formula-backed charts (status column, category bar, full-width stacked site
bar) with finding-led titles linked to Dashboard_Summaries; slicers for
Site, Job_Category, Status and Assigned_To that filter every tile and chart
through SUBTOTAL visibility. Use structured references, not fixed 120-row
ranges, and never merge cells. Baseline check at 2026-09-30 All:120
total,48 Closed,12 Cancelled,60 backlog,50 overdue,30/48 on-time,35827.50
cost. Verify site/no-Closed/no-row tests, source preservation and a
separate two-row append sandbox with 122 total,49 Closed,61 backlog,51
overdue,31/49 on-time and 36227.50 cost. These are snapshot tests, not
universal future figures. Ask for missing inputs; never approve
assignments, infer causes/savings or send email. Return the entire file.
```

2. Use Excel Settings (...) > Manage plugins & skills > Custom skills > Open skills folder (Create OneDrive folder if needed). Copy the reviewed water-service-dashboard folder into that app-created location, Refresh, enable and verify its exact name in Choose plugins & skills. If unavailable, paste the entire reviewed Skill before each prompt, remove @water-service-dashboard and record Manual practice - native Skill unavailable. Installing maintenance-reporting or water-service-pareto does not install this distinct Skill.

3. Open data/templates/Northstar_Dashboard_Starter.xlsx and save as Northstar_Dashboard.xlsx in OneDrive. If continuing your own workbook, copy its CLEAN tblWorkOrders into the starter, keeping exact headers and table name; do not create a second table. The starter has a labelled blank Dashboard and four selection cells B2:B5, initially All.

EXCEL - CHAT ONLY:

```
@water-service-dashboard Inspect this Northstar Water Services dashboard
starter. Confirm tblWorkOrders, 120 unique IDs, reporting date 2026-09-30
and selection cells B2:B5 initially All. Explain which source and control
sheets must remain intact. Report discrepancies before any edits.
```

4. Read Control and confirm the fixed reporting date. Ask for a visual plan before edits. Keep source, Checks and Prompt_Log separate from the Dashboard view. A dashboard is a decision interface: each visual should answer a maintenance question rather than decorate the page.

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```
EXCEL - PLAN:

@water-service-dashboard Plan Dashboard for Northstar maintenance at 2026-
09-30. Use selection cells B2:B5 for Site, Job_Category, Status,
Assigned_To, all initially All. Reserve B7 total orders, B8 backlog, B9
overdue backlog, B10 closed, B11 on-time closure rate, B12 Cost_SGD. Place
three charts for status, category and site with labels, units and scope.
Preserve source data.
```

5. Ask Copilot to insert formulas for the six measures using structured table references. For now all selectors are All; in Lab 8 the same formulas must honour the four filters. Verify the denominator and use N/A, not 0%, if no Closed rows remain.

```
EXCEL - ALLOW EDITING:

@water-service-dashboard Build the planned six KPI formulas from
tblWorkOrders. Backlog excludes Closed/Cancelled. Overdue also requires
Due_Date < Control!B2. On-time rate counts Closed_Date <= Due_Date among
Closed rows only, and shows N/A when none. Recorded cost sums Cost_SGD of
all included statuses. Link to table columns, never a fixed 120-row range.
Keep source data unchanged.
```

6. Add formula-backed chart summaries for the six statuses, five categories and five sites. Use bar/column charts: category and site are nominal groups; a line chart would imply an order that is not present. Label count axes and avoid 3-D effects. Then polish the charts with the built-in @chart-design skill from Lab 5.

```
EXCEL - ALLOW EDITING:

@water-service-dashboard Create three editable bar or column charts using
formula-backed summaries: work-order count by Status, by Job_Category and
by Site. Each summary must reconcile to the total selected orders. Use
white background, navy titles, blue/teal accents, readable labels and a
visible reporting date. Do not use category counts as evidence of causes or
use colour alone for status.
```

7. In Checks confirm sum of each chart's categories equals 120 with All filters. Compare six KPIs to Control and record the on-time numerator/denominator. Add Evidence_Log entries for each chart and metric. If Copilot has used hard-coded numbers, replace them with formulas before Lab 8.

```
EXCEL - CHAT ONLY:

@water-service-dashboard Reconcile the Northstar Water Services dashboard
with Control at All selections. Compare all six KPIs and totals of the
Status, Job_Category and Site charts. Flag hard-coded values, fixed row
ranges, inconsistent scopes and any interpretation that treats job-category
frequency as a cause. Do not edit.
```

8. Create a measure dictionary on a new Measures sheet: metric, business definition, formula, source fields, date rule, scope and limitation. Ask Copilot to

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explain formulas, then read one overdue example and one closed example from the source table yourself. Save the baseline dashboard with all filters All.

```
EXCEL - CHAT ONLY:

@water-service-dashboard Explain each Dashboard measure in plain English,
including terminal statuses, strict overdue date rule, closed-only SLA
denominator and meaning of recorded cost. Give one source-row example for
overdue and one for on-time closure; cite Work_Order_ID. Do not guess a
root cause or financial saving.
```

Sample prompts

Purpose	Prompt
Scope	State actual source, report date and selected scope before acting.
Verify	Explain the source/formula checks and preserve unknowns and original records.

Test it

Check	Expected result
Baseline	Total 120; backlog 60; overdue 50; closed 48.
SLA and cost	30/48 = 62.5%; S$35,827.50.
Chart reconciliation	Three count summaries each total 120; formulas reference tblWorkOrders.
Decision design	Measures dictionary and source/date visible; no unsupported cause claims.

Checkpoint and rejoin point

If you fell behind, use data/checkpoints/Northstar_Verified_Checkpoint.xlsx and data/checkpoints/Northstar_Verified_Snapshot.docx. Record checkpoint used in Prompt_Log; then perform this lab, rather than claiming the checkpoint as your output.

Troubleshooting

If this happens	Fix
Copilot cannot find the source	Use the app file picker to select the actual OneDrive file. A typed local path or a different app chat is not shared context. If needed copy only the verified snapshot into the current app.
A total disagrees with Control	Clear dashboard filters to All, check table extent and duplicate IDs, recalculate, then compare definitions and reporting date. Reconcile before continuing.
The named feature is missing	Confirm the correct assigned account, English display language and app version. Use the stated manual practice route; record feature unavailable. This is not evidence that a native skill ran.

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Challenge

Repeat the task for one different site and explain how the filter scope changes the result. Keep the original verified output.

Reflection

Which evidence proves the output is correct, and which decision still needs a person?

Note: The lab folder labs/lab-07-*/ holds the instruction PDF with sample prompts, a data folder with templates, checkpoints and the mock-data workbook (data/Northstar_Work_Orders.xlsx), plus data/templates/water-service-dashboard/SKILL.md (Reviewed interactive dashboard Skill fallback). Northstar Water Services is a fictional water-pipe maintenance and service company. All service hubs, staff and work orders are synthetic. Use the fixed reporting date 2026-09-30; missing facts remain UNKNOWN. Proposals do not change real assignments. Training email drafts are never sent.

Concepts for Lab 8: One Filter Scope for Every Visual, Refresh Tests and Static Snapshots.

One Filter Scope for Every Visual

Interactive means a learner-controlled selection changes the evidence shown. This dashboard has two controls that work together: four dropdowns in B2:B5 and four table slicers. All bypasses one dimension only; selections combine with AND. Every KPI tile and chart summary multiplies the dropdown mask by SUBTOTAL row visibility, so a slicer that hides source rows changes every measure too.

A dashboard can appear interactive while only one chart changes. Test a single-site selection with the dropdown and with the slicer, a combined category/owner selection and an empty selection. Check the scope line and the D3 slicer banner, and compare every chart total with the selected total.

How it works

- Define the selector cells.
- Apply the same AND mask everywhere.
- Test each control and edge case.
- Reconcile selected chart and KPI totals.

Worked example

- Woodlands Hub in B2 or in the Site slicer gives 24 work orders.
- Status New gives zero Closed and an N/A closure rate.
- D3 warns while any slicer hides source rows.

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Decision guide

Use when	Avoid when
All output calculations follow the same scope.	A selector only changes a decorative title or one chart.

Authoritative references

- https://support.microsoft.com/en-us/excel/get-started/use-slicers-to-filter-data
- https://support.microsoft.com/en-us/excel/create-and-share-a-dashboard-with-excel-and-microsoft-groups

Refresh Tests and Static Snapshots

A refresh test deliberately introduces known new records and verifies the expected delta. Table references should expand when rows are appended; PivotTables additionally need a refresh and valid source extent. The course adds two rows in a sandbox copy so the live baseline remains 120 rows. A Word/PDF/PowerPoint snapshot freezes evidence and does not automatically update when the Excel dashboard changes.

A correct dashboard today may fail after the next export if formulas reference a fixed cell range. Known delta testing reveals that failure. Preserving a separate baseline prevents a test record from contaminating the management pack. A snapshot should record source, as-at date, filters and version so its claims remain interpretable after the workbook is refreshed.

How it works

- Save an isolated sandbox copy.
- Append known unique test rows.
- Recalculate and verify expected deltas.
- Handoff the untouched baseline.

Worked example

- Two test rows increase total by2 and recorded cost byS$400.
- Only one new row adds to backlog/overdue; the other adds to on-time Closed.
- The final deck uses the untouched 120-row baseline, not the test copy.

Decision guide

Use when	Avoid when
A predictable change can be checked independently.	Test rows are being mixed into the report without disclosure.

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Authoritative references

- https://support.microsoft.com/en-us/excel/create-and-share-a-dashboard-with-excel-and-microsoft-groups

Lab 8 · Interactive Filters and Refresh — Use the Dashboard Skill for Filters, Refresh and Safe Reruns

Learning outcome: LO3: build and validate an interactive Excel maintenance dashboard.

Goal: Continue Lab 7 by making four dropdown selectors control every KPI and chart. Test a single site, a zero-denominator selection and a combined category/owner filter. Then append two synthetic work orders in a sandbox copy and prove the dashboard expands. The dashboard also has four table slicers; every tile and chart follows the dropdowns and the slicers together.

Continue Lab 7 by making four dropdown selectors control every KPI and chart. Test a single site, a zero-denominator selection and a combined category/owner filter. Then append two synthetic work orders in a sandbox copy and prove the dashboard expands. The dashboard also has four table slicers; every tile and chart follows the dropdowns and the slicers together.

What you'll build

Northstar_Dashboard.xlsx with four working filters plus Northstar_Dashboard_Refresh_Test.xlsx, filter evidence and a verified baseline snapshot for PowerPoint. (Tools: Excel Copilot · Microsoft 365 Copilot web · OneDrive.)

Prerequisites

- Microsoft 365 account with Copilot available in Word, Excel, PowerPoint and Outlook. Availability depends on the assigned licence and tenant settings; confirm access with the trainer before class.
- Desktop Excel on Windows or Mac is recommended; save working copies to your assigned OneDrive course folder with AutoSave on. Do not overwrite the supplied input.
- Download this whole lab folder. All files are synthetic; work-genai is your own learner output folder.
- Lab 7 dashboard or supplied formula checkpoint.
- Lab 7 reviewed water-service-dashboard Skill installed or explicitly labelled manual route.

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Step-by-step

1. Open your Lab 7 baseline dashboard. Rejoin using data/checkpoints/Northstar_Verified_Checkpoint.xlsx only if needed; record that it is an example formula dashboard. Set all four selections B2:B5 to All and confirm 120. Save baseline before creating any refresh-test copy.

```
EXCEL - CHAT ONLY:

@water-service-dashboard Before testing filters, report the baseline
Northstar Water Services dashboard source, date and scope. Confirm 120
unique work orders and all four selectors All. Identify the baseline file
that must remain untouched when making the refresh-test copy.
```

2. Use Data > Data Validation > Allow List to set dropdowns on B2 Site, B3 Job_Category, B4 Status and B5 Assigned_To. Include All plus the source values, using Filter_Lists if supplied. Ask Copilot to apply all four selections to EVERY KPI and chart summary. A selection of All bypasses only that filter; selected dimensions combine with AND.

```
EXCEL - PLAN:

@water-service-dashboard Make Dashboard B2:B5 control all six KPIs and
three chart summaries with AND logic. All bypasses its dimension. Use
COUNTIFS/SUMIFS or SUMPRODUCT with structured table references. Blank owner
means unassigned and must not disappear from All. Keep N/A for no closed
jobs.



EXCEL - ALLOW EDITING:

@water-service-dashboard Apply the reviewed filter plan and dropdown lists.
Keep existing source rows and chart positions. Show the selected scope
beside the reporting date and make all charts recalculate from the same
selected rows.
```

3. Choose Woodlands Hub in B2 and leave other selectors All. Then set B2 back to All and click Woodlands Hub in the Site slicer instead: the D3 banner reports 24 of 120 rows and every tile must show the same values. Clear the slicer afterwards with its Clear Filter button. Record evidence in the Evidence_Log sheet.

```
EXCEL - CHAT ONLY:

@water-service-dashboard For the dashboard selection Site = Woodlands Hub
and all other dimensions All, list the expected six KPI values from
tblWorkOrders and verify each chart sum matches the selected total. Confirm
the Site slicer gives the same six values and that D3 reports the slicer
filter.
```

4. Set Status to New with other selectors All. Confirm zero closed -> N/A. Then set Site = Woodlands Hub, Job_Category = Leak Repair, Assigned_To = Daniel Lee,

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Status = All, a combination with no rows; total, backlog, overdue, closed and cost must be 0 and rate N/A. Reset all four to All and verify baseline controls. Repeat the Status = New case with the Status slicer instead of B4 and confirm the same N/A; then clear every slicer.

```
EXCEL - CHAT ONLY:

@water-service-dashboard Check two dashboard edge cases without changing
source data: Status New with other selectors All; then Site Woodlands Hub,
Job_Category Leak Repair, Assigned_To Daniel Lee, Status All. Verify zero
closed gives N/A and no selected rows gives zero count/cost with N/A rate.
Show how all charts follow the same selection. Restore all selectors to
All.
```

5. Save a COPY as Northstar_Dashboard_Refresh_Test.xlsx. In the last row of tblWorkOrders press Tab to append, then copy the two rows from the Refresh_Test_Rows sheet (they already hold real Excel dates). WO-2026-0121 is Assigned, due 2026-09-29 and cost 150; WO-2026-0122 is Closed on time and cost 250. No existing IDs are edited.

6. Recalculate with all selections All. If using PivotTables also Data > Refresh All and inspect source extent. Verify total122, closed49, backlog61, overdue baseline+1, cost baseline+400 and on-time numerator baseline+1 over49. If formulas do not expand, fix structured references/table extent. Keep this sandbox copy as refresh evidence; do not use its figures in the baseline presentation.

```
EXCEL - CHAT ONLY:

@water-service-dashboard Verify the sandbox append test for the two new
work orders. Reconcile total 122, closed 49, backlog 61, overdue increased
by 1, cost increased by S$400 and on-time numerator increased by 1. Confirm
no original source record changed and all three charts total 122 with All
selections. Compare original IDs, statuses, owners, dates and costs to the
saved baseline; all original 120 rows must match.
```

7. In the sandbox workbook request a controlled Skill rerun after the append. Review the named replacement areas, then approve only those areas. Verify one dashboard, three charts, original 120 source rows unchanged and both legitimate new rows retained. Keep the 120-row baseline separate for Lab 9.

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```
EXCEL - PLAN:

@water-service-dashboard Refresh the existing dashboard for the sandbox
122-row table and same reporting date. Preserve all original source
records, both new rows, .Rules, Pareto_* and unrelated sheets. Identify
named output replacements and ask before editing; keep one dashboard and
three charts.


EXCEL - ALLOW EDITING:

@water-service-dashboard Apply only the reviewed sandbox dashboard rerun.
Retain the chosen source/date and make every KPI and chart follow the four
filters; reconcile 122 total,61 backlog,51 overdue,49 Closed,31/49 on-time
and S$36,227.50. Do not alter the untouched baseline workbook.
```

8. Return to the untouched 120-row Northstar_Dashboard.xlsx. Set selectors All and record a frozen evidence snapshot dated 2026-09-30 for Lab 9. Copy data/templates/Northstar_Operations_Brief_Template.docx to Northstar_Verified_Snapshot.docx in Word and insert the reviewed summary and chart values. Save chart images if needed with right-click chart > Save as Picture; the slide copy will be a static snapshot, not a live dashboard.

```
WORD COPILOT:

Using only the verified baseline dashboard evidence I paste here, prepare
the frozen Northstar Water Services presentation snapshot. Include
reporting date, All scope, 120-row baseline, six KPIs, chart values, source
workbook link and limitations. Exclude the 122-row sandbox. Keep proposed
priorities separate from observed facts.
```

Sample prompts

Purpose	Prompt
Scope	State actual source, report date and selected scope before acting.
Verify	Explain the source/formula checks and preserve unknowns and original records.

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Test it

Check	Expected result
Woodlands control	Total 24; backlog 8; overdue 7; closed 16; cost S$6,668.50; on-time 10/16.
Interactive scope	Changing any dropdown or slicer affects all six KPI tiles and all chart counts; D3 reports slicer filters.
Edge case	No Closed -> N/A; no rows -> zero count/cost and N/A rate.
Refresh sandbox	122 total;49 closed;61 backlog;51 overdue;S$36,227.50;31/49 on-time.
Handoff baseline	Return to 120-row untouched baseline; date and All scope recorded.

Checkpoint and rejoin point

If you fell behind, use data/checkpoints/Northstar_Verified_Checkpoint.xlsx and data/checkpoints/Northstar_Verified_Snapshot.docx. Record checkpoint used in Prompt_Log; then perform this lab, rather than claiming the checkpoint as your output.

Troubleshooting

If this happens	Fix
Copilot cannot find the source	Use the app file picker to select the actual OneDrive file. A typed local path or a different app chat is not shared context. If needed copy only the verified snapshot into the current app.
A total disagrees with Control	Clear dashboard filters to All, check table extent and duplicate IDs, recalculate, then compare definitions and reporting date. Reconcile before continuing.
The named feature is missing	Confirm the correct assigned account, English display language and app version. Use the stated manual practice route; record feature unavailable. This is not evidence that a native skill ran.

Challenge

Repeat the task for one different site and explain how the filter scope changes the result. Keep the original verified output.

Reflection

Which evidence proves the output is correct, and which decision still needs a person?

Note: The lab folder labs/lab-08-*/ holds the instruction PDF with sample prompts, a data folder with templates, checkpoints and the mock-data workbook (data/Northstar_Work_Orders.xlsx), plus data/templates/water-service-dashboard/SKILL.md (Reviewed dashboard Skill for reruns). Northstar Water Services is a fictional water-pipe maintenance and service company. All service hubs, staff and work orders are synthetic. Use the fixed reporting date 2026-09-30; missing facts

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remain UNKNOWN. Proposals do not change real assignments. Training email drafts are never sent.

Topic 3 Recap - Mapped to Learning Outcomes

Use this checkpoint to explain what you can now do and identify any lab evidence you still need to repair.

Learning outcome	Evidence from this topic
LO3	Six KPI tiles and three charts follow four dropdowns and slicers.
Verify	Baseline controls and known refresh deltas.

Topic 4 — Create a Professional Presentation with PowerPoint Copilot

Continue the dashboard into a decision pack

Key concepts

- Evidence story — Observation, implication, action and decision.
- Charts — Reviewed values, provenance and snapshot date.
- Refinement — Hierarchy, notes and actual rehearsal feedback.
- Handoff — Reconciled pack and unsent communication.

Concepts for Lab 9: From Dashboard Evidence to a Decision Story, Charts, Templates and Source Provenance.

From Dashboard Evidence to a Decision Story

A management presentation transforms evidence into a sequence of observations, implications, proposed actions and a decision request. Each slide needs one defensible headline and a register entry linking the claim to its measure, source, reporting date and filter scope. A chart contains data; the headline interprets it. That interpretation must stay within what the data supports.

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Copilot can accelerate outlines and drafting, but a persuasive invented conclusion is still wrong. Before generating the deck, use Word to organise an eight-slide brief with a supporting chart and limitation for each claim. This explicit bridge makes the Excel-to-PowerPoint handoff reliable and avoids assuming a workbook can be used by every presentation-generation mode.

How it works

- Freeze the verified dashboard snapshot.
- Register each proposed claim.
- Write a reviewed Word brief.
- Generate and reconcile the deck.

Worked example

- Backlog by site supports prioritisation, not a proven cause.
- Available-hours evidence supports a capacity discussion, not confirmed allocation.
- The final slide asks management to decide on a proposed plan.

Decision guide

Use when	Avoid when
Every headline has a checked measure and clear scope.	The story introduces savings, causes or approvals absent from evidence.

Authoritative references

- https://support.microsoft.com/en-gb/word/copilot/draft-and-add-content-with-copilot-in-word
- https://support.microsoft.com/en-us/powerpoint/copilot/add-a-slide-from-a-file-with-copilot-in-powerpoint
- https://support.microsoft.com/en-us/microsoft-365-copilot/file-formats-supported-by-microsoft-365-copilot

Charts, Templates and Source Provenance

A template supplies slide layouts, typography and colours; it does not supply correct facts. PowerPoint Copilot can draft slides from selected supported files. Reviewed Word briefs are a reliable narrative input for this course. Excel charts may be embedded for editing or pasted as images; both are copies of a reporting state unless a separate supported link is deliberately configured and tested.

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Approximate AI-generated charts can contradict source values while looking plausible. Replace them with reviewed charts or verified tables and keep source/date notes. Distinguish all-work-order counts from backlog-only counts. A professional presentation uses consistent units and chart semantics, not just a consistent palette.

How it works

- Select the reviewed brief.
- Apply the existing template.
- Insert verified charts or tables.
- Record source and snapshot scope.

Worked example

- An editable embedded chart remains a copy rather than a tested live link.
- An image is labelled as a snapshot at 30 September.
- All eight slides use one reviewed baseline.

Decision guide

Use when	Avoid when
The chart object or image can be verified against the register.	A visual is assumed accurate because Copilot produced it.

Authoritative references

- https://support.microsoft.com/en-us/powerpoint/copilot/add-a-slide-from-a-file-with-copilot-in-powerpoint
- https://support.microsoft.com/en-us/microsoft-365-copilot/file-formats-supported-by-microsoft-365-copilot

Lab 9 · Dashboard to presentation — Turn Dashboard Evidence into a Management Story

Learning outcome: LO4: turn verified dashboard evidence into a professional management presentation.

Goal: Continue the baseline dashboard from Topic 3. Build a slide evidence register and a Word presentation brief before asking PowerPoint Copilot for an eight-slide management deck. The story connects work-order status, site backlog, service delivery and assignment proposals to a specific decision; it does not introduce unsupported savings or causes.

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Continue the baseline dashboard from Topic 3. Build a slide evidence register and a Word presentation brief before asking PowerPoint Copilot for an eight-slide management deck. The story connects work-order status, site backlog, service delivery and assignment proposals to a specific decision; it does not introduce unsupported savings or causes.

What you'll build

Northstar_Presentation_Brief.docx, Northstar_Management_Draft.pptx and a slide-by-slide evidence register tied to the 120-row dashboard snapshot. (Tools: Microsoft 365 Copilot · Excel · Word · PowerPoint · Outlook · OneDrive.)

Prerequisites

- Microsoft 365 account with Copilot available in Word, Excel, PowerPoint and Outlook. Availability depends on the assigned licence and tenant settings; confirm access with the trainer before class.
- Desktop Excel on Windows or Mac is recommended; save working copies to your assigned OneDrive course folder with AutoSave on. Do not overwrite the supplied input.
- Download this whole lab folder. All files are synthetic; work-genai is your own learner output folder.
- Verified Lab 8 baseline dashboard and frozen evidence snapshot.

Step-by-step

1. Open the untouched baseline Northstar_Dashboard.xlsx from Lab 8; confirm All selectors, total120 and report date2026-09-30. Do not use the 122-row sandbox. If needed use this lab's checkpoint and record rejoin. Record verified metric/chart values in the Evidence_Log sheet with source sheet/table, scope and date.

```
EXCEL - CHAT ONLY:

For the untouched 120-row Northstar Water Services baseline dashboard at
2026-09-30, All selectors, return a presentation evidence table: six KPIs,
backlog by site, status and job-category counts, unassigned backlog and
relevant source references. Do not use the refresh-test copy or infer
service disruption, safety risk, causes or savings.
```

2. Open data/templates/Northstar_Presentation_Brief_Template.docx in Word. Add the verified summary and chart values plus the OneDrive workbook link. Save as Northstar_Presentation_Brief.docx. Copilot must see selected files or pasted evidence; the link alone and the Excel conversation do not guarantee accessible context.

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```
WORD COPILOT:

Using this verified 120-row Northstar maintenance snapshot, draft an eight-
slide management presentation brief. For each slide give one claim
headline, supporting measure/chart, source/date/scope, limitation and
speaker note. Structure: overview, portfolio, site backlog, status/category
mix, SLA/overdue, assignment/capacity, proposed actions/trade-offs,
decision/next review. Do not invent causes, savings, approval or future
dates; use UNKNOWN when absent.
```

3. Review the brief against source values. A bar showing counts by site is not a severity index; if you want backlog by site, count only nonterminal statuses. Correct any misleading headline and fill the slide evidence register. Suggested actions must cite a finding and say proposed.

```
WORD COPILOT:

Review this eight-slide Northstar Water Services brief against the supplied
baseline evidence. Check each headline has a supporting value, correct
units, source/date/scope and an explicit limitation. Rewrite unsupported
claims as questions or proposed actions. Do not change verified figures or
invent approvals.
```

4. Open the supplied PowerPoint template, save as Northstar_Management_Draft.pptx and open Copilot. Choose Create presentation from file if offered, or reference the Word brief while editing the existing deck. Select your reviewed OneDrive Word brief; direct workbook-to-deck creation is not assumed. Invoke @maintenance-reporting if available.

```
POWERPOINT COPILOT:

@maintenance-reporting Create or update an eight-slide management
presentation from the selected Northstar_Presentation_Brief.docx. Follow
the existing white/navy/blue template. One evidence-based message per
slide, short visible text, charts or tables for numerical evidence,
source/as-at/scope in notes, and a final decision request. Keep values
exact, and label all next actions as proposed. Do not add extra slides or
invented facts.
```

5. Replace approximate or invented charts with your reviewed Excel charts: copy chart in Excel, then PowerPoint Home > Paste > Paste Special > Microsoft Excel Chart Object for an editable embedded chart when supported; otherwise paste a static image and label it snapshot. Embedded copies and images do not automatically follow the original workbook. Use direct verified values if a chart object is unsupported.

6. Inspect all eight slides and notes. Check chart title, axis, unit, status definition, date and scope. Each numeric headline must match one register row. Confirm the deck uses baseline120, not refresh122. Save for Lab 10; do not represent the draft as an approved plan.

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```
POWERPOINT COPILOT:

Audit the eight-slide Northstar Water Services management draft against the
selected presentation brief and evidence register. Confirm baseline 120,
source notes, chart labels and units, correct backlog exclusions and
proposed-action wording. List needed corrections and preserve slide count;
do not claim Excel charts update automatically.
```

Sample prompts

Purpose	Prompt
Scope	State actual source, report date and selected scope before acting.
Verify	Explain the source/formula checks and preserve unknowns and original records.

Test it

Check	Expected result
Evidence continuity	Every deck figure maps to the 120-row Lab 8 baseline and slide evidence register.
Eight-slide narrative	Observation -> implication -> proposed action -> decision; no invented cause/savings.
Chart semantics	Backlog charts exclude Closed/Cancelled; embedded/image copies labelled snapshot.

Checkpoint and rejoin point

If you fell behind, use data/checkpoints/Northstar_Verified_Checkpoint.xlsx and data/checkpoints/Northstar_Verified_Snapshot.docx. Record checkpoint used in Prompt_Log; then perform this lab, rather than claiming the checkpoint as your output.

Troubleshooting

If this happens	Fix
Copilot cannot find the source	Use the app file picker to select the actual OneDrive file. A typed local path or a different app chat is not shared context. If needed copy only the verified snapshot into the current app.
A total disagrees with Control	Clear dashboard filters to All, check table extent and duplicate IDs, recalculate, then compare definitions and reporting date. Reconcile before continuing.
The named feature is missing	Confirm the correct assigned account, English display language and app version. Use the stated manual practice route; record feature unavailable. This is not evidence that a native skill ran.

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Challenge

Repeat the task for one different site and explain how the filter scope changes the result. Keep the original verified output.

Reflection

Which evidence proves the output is correct, and which decision still needs a person?

Note: The lab folder labs/lab-09-*/ holds the instruction PDF with sample prompts, a data folder with templates, checkpoints and the mock-data workbook (data/Northstar_Work_Orders.xlsx). Northstar Water Services is a fictional water-pipe maintenance and service company. All service hubs, staff and work orders are synthetic. Use the fixed reporting date 2026-09-30; missing facts remain UNKNOWN. Proposals do not change real assignments. Training email drafts are never sent.

Concepts for Lab 10: Professional Refinement and Rehearsal, Reviewed Handoff and Artifact Lifecycle.

Professional Refinement and Rehearsal

Refinement improves the clarity of the existing evidence without changing its meaning. Claim-led titles, short visible text, readable axes, consistent chart styles and concise speaker notes help an audience understand the maintenance decision. Rehearsal tests whether the audience can identify the reporting scope, the key finding and the decision required. It produces observed feedback, not a fabricated checklist completion.

A well-designed deck can still fail when the speaker’s notes contain different numbers or a proposal is spoken as an approved action. Review notes as part of the evidence pack and inspect the exported PDF at presentation size. Exporting can alter layout or font rendering, so the PDF must be opened and checked rather than assumed identical to the source.

How it works

- Audit the draft against evidence.
- Apply reviewed design changes.
- Rehearse and record actual feedback.
- Inspect PPTX and exported PDF.

Worked example

- One message per slide; chart labels remain readable.
- A 45-second note explains evidence and limitation.
- Partner feedback leads to one recorded revision.

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Decision guide

Use when	Avoid when
The message and decision can be understood quickly.	Polish hides an unsupported claim or unreadable chart.

Authoritative references

- https://support.microsoft.com/en-us/powerpoint/copilot/copilot-in-powerpoint-skills

Reviewed Handoff and Artifact Lifecycle

The final pack contains an interactive Excel dashboard, a fixed reporting brief, a static PowerPoint/PDF snapshot, the custom skill and a reviewed communication draft. These artifacts have different lifecycles. The dashboard may refresh; the skill defines a method; the presentation records a dated claim set. Link each file accurately and preserve the evidence register so a later reader knows what was reviewed.

Updating one artifact without the others creates inconsistent communication. Reconcile the final pack and explicitly state that actions remain proposed and the email remains a draft. Versioned source and archived superseded artifacts let the team revisit an earlier report without confusing it with the current one. A missing link is UNKNOWN until an actual accessible link is supplied.

How it works

- Reconcile final artifacts.
- Record actual filenames and links.
- Review the handoff draft.
- Preserve the dated snapshot and skill.

Worked example

- PowerPoint and PDF both show eight slides/pages from the same baseline.
- Outlook does not claim a missing attachment was sent.
- The next update reruns the method with new verified inputs.

Decision guide

Use when	Avoid when
Current versions and review evidence are explicit.	A refreshed dashboard is assumed to rewrite an old presentation.

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Authoritative references

- https://support.microsoft.com/en-us/microsoft-365-copilot/file-formats-supported-by-microsoft-365-copilot
- https://support.microsoft.com/en-us/powerpoint/copilot/copilot-in-powerpoint-skills

Lab 10 · Professional handoff — Refine and Present the Maintenance Decision Deck

Learning outcome: LO4: turn verified dashboard evidence into a professional management presentation.

Goal: Refine the management deck using the same custom skill and reviewed dashboard evidence. Improve hierarchy, consistency and speaker notes, test the story in a short rehearsal, and prepare a reviewed Outlook handoff draft. The final pack preserves the distinction between a static presentation snapshot, a refreshable dashboard and a manager's unmade decision.

Refine the management deck using the same custom skill and reviewed dashboard evidence. Improve hierarchy, consistency and speaker notes, test the story in a short rehearsal, and prepare a reviewed Outlook handoff draft. The final pack preserves the distinction between a static presentation snapshot, a refreshable dashboard and a manager's unmade decision.

What you'll build

Northstar_Management_Final.pptx/PDF, a rehearsal record, final evidence register and unsent Outlook handoff draft with accurate artifact links. (Tools: Microsoft 365 Copilot · Excel · Word · PowerPoint · Outlook · OneDrive.)

Prerequisites

- Microsoft 365 account with Copilot available in Word, Excel, PowerPoint and Outlook. Availability depends on the assigned licence and tenant settings; confirm access with the trainer before class.
- Desktop Excel on Windows or Mac is recommended; save working copies to your assigned OneDrive course folder with AutoSave on. Do not overwrite the supplied input.
- Download this whole lab folder. All files are synthetic; work-genai is your own learner output folder.
- Lab 9 management draft, reviewed presentation brief and evidence register.

Step-by-step

1. Open your Lab 9 draft and save as Northstar_Management_Final.pptx. Rejoin only if necessary using data/checkpoints/Northstar_Management_Example.pptx and

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snapshot; review and adapt its supplied example charts and narrative against the verified snapshot. Open the same reviewed brief in the file picker; confirm native skill or label manual practice.

2. Ask the skill to audit before editing. It should report crowded slides, weak evidence, missing labels and inconsistent scope. Review the list yourself against the evidence register; styling changes must never change values.

```
POWERPOINT COPILOT:

@maintenance-reporting Audit this eight-slide deck against the selected
verified Northstar brief. List inaccurate figures, ambiguous units,
date/scope omissions, unsupported causal or approval wording, dense text
and weak hierarchy. Propose corrections; do not change values or source
facts.
```

3. Apply only the reviewed corrections. Keep white/navy/blue with teal highlights, one main message, readable labels, consistent chart formats and source notes. Ask for concise speaker notes; check that notes do not introduce new numbers or promise assignment approvals.

```
POWERPOINT COPILOT:

@maintenance-reporting Apply the reviewed design and wording corrections to
the existing eight slides. Improve hierarchy and whitespace, make titles
claim-led, retain accurate figures and charts, and add 45-second speaker
notes per slide. Separate observed findings from proposed actions and end
with the decision requested. Preserve the baseline date/scope; do not
duplicate slides.
```

4. Use Slide Show to inspect at presentation size. Run a five-minute rehearsal with a partner or PowerPoint Rehearse with Coach if available. Ask the partner to identify the reporting date, highest-priority evidence and decision requested. Record one actual change in the Evidence_Log sheet; do not fabricate feedback.

```
POWERPOINT COPILOT:

Using only this Northstar Water Services decision deck, propose three
rehearsal questions a service operations manager could ask about overdue
jobs, technician capacity and proposed priorities. Suggest where the
existing evidence supports an answer and where it remains UNKNOWN. Do not
invent partner feedback or alter the evidence.
```

5. Export File > Export or Save As > PDF. Open the PDF and verify eight pages, readable charts and all visible values. Compare PPTX, PDF, Word brief and dashboard against the evidence register. Keep the source workbook with All filters and the archived refresh-test copy separately.

6. Open Outlook and create a draft to operations.manager@example.invalid. Provide the reviewed brief and the actual OneDrive links to the dashboard/deck. Ask Copilot to draft a handoff email; it may not invent shared links or claim a sent attachment. Keep the email unsent and save a local reviewed TXT copy.

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OUTLOOK COPILOT:

```
Draft a short Northstar management handoff email using this verified brief
and the links I provide. State the 30 September 2026 baseline scope,
headline findings, proposed actions and decision required. Explain that the
PowerPoint/PDF is a static snapshot while the Excel dashboard is
interactive. Leave missing links UNKNOWN. Do not claim approval or send the
draft.
```

7. Complete a final handoff checklist in the Evidence_Log sheet: workbook source and controls, working filters, sandbox refresh proof, skillv2 installation/manual record, eight-slide PPTX/PDF, reviewed notes, evidence register and unsent draft. Record the exact filenames and your actual checks. Save the final pack in work-genai.

WORD COPILOT:

```
Review the final Northstar Water Services handoff evidence I supply:
baseline workbook, refresh-test record, skill v2, eight-slide deck and PDF,
brief, source register and unsent email draft. Return a completeness
checklist with verified evidence and missing items UNKNOWN. Confirm every
artifact uses the same baseline date/scope; never claim approval,
submission or sending.
```

Sample prompts

Purpose	Prompt
Scope	State actual source, report date and selected scope before acting.
Verify	Explain the source/formula checks and preserve unknowns and original records.

Test it

Check	Expected result
Presentation integrity	Eight slides and eight PDF pages; visible figures agree with verified register.
Professional delivery	Readable chart labels, consistent design, evidence-specific notes and actual rehearsal improvement.
Final continuity	Baseline dashboard, brief, deck and unsent draft agree; static snapshot stated.
Decision boundary	No real email sent or work order reassigned; proposed actions remain proposed.

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Checkpoint and rejoin point

If you fell behind, use data/checkpoints/Northstar_Verified_Checkpoint.xlsx and data/checkpoints/Northstar_Verified_Snapshot.docx. Record checkpoint used in Prompt_Log; then perform this lab, rather than claiming the checkpoint as your output.

Troubleshooting

If this happens	Fix
Copilot cannot find the source	Use the app file picker to select the actual OneDrive file. A typed local path or a different app chat is not shared context. If needed copy only the verified snapshot into the current app.
A total disagrees with Control	Clear dashboard filters to All, check table extent and duplicate IDs, recalculate, then compare definitions and reporting date. Reconcile before continuing.
The named feature is missing	Confirm the correct assigned account, English display language and app version. Use the stated manual practice route; record feature unavailable. This is not evidence that a native skill ran.

Challenge

Repeat the task for one different site and explain how the filter scope changes the result. Keep the original verified output.

Reflection

Which evidence proves the output is correct, and which decision still needs a person?

Note: The lab folder labs/lab-10-*/ holds the instruction PDF with sample prompts, a data folder with templates, checkpoints and the mock-data workbook (data/Northstar_Work_Orders.xlsx). Northstar Water Services is a fictional water-pipe maintenance and service company. All service hubs, staff and work orders are synthetic. Use the fixed reporting date 2026-09-30; missing facts remain UNKNOWN. Proposals do not change real assignments. Training email drafts are never sent.

Topic 4 Recap - Mapped to Learning Outcomes

Use this checkpoint to explain what you can now do and identify any lab evidence you still need to repair.

Learning outcome	Evidence from this topic
LO4	Eight-slide evidence story and reviewed handoff.
Verify	PDF, notes, register and actual rehearsal agree.

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Topic 5 — Copilot Agents and Mini Capstone

Build a work-order agent, then apply the whole workflow in groups

Key concepts

- Agent Builder — Create a specialist Copilot agent without code.
- Knowledge — OneDrive files the agent searches and cites.
- Instructions & Skills — Role, rules and a reusable work-order Q&A method.
- Mini capstone — Groups turn safe data into a dashboard and decision deck.

Concepts for Lab 11: What Is Agent Builder?.

What Is Agent Builder?

Agent Builder is the no-code tool in Microsoft Copilot for creating your own specialist agent. An agent is a focused version of Copilot with its own name, instructions, knowledge, skills, capabilities and starter prompts. You open it from Agents & Skills > New agent, describe the agent in plain English or fill in Configure, test it in Try it, then create and share it.

An agent packages a recurring job, here answering work-order questions, so colleagues get consistent, grounded and bounded answers without rewriting prompts. Its answers are only as good as its instructions and knowledge, and it inherits the same permission and licensing rules as Copilot. Skills in agents are a preview that cannot yet be combined with uploaded files.

How it works

- Describe the agent’s job.
- Configure instructions and knowledge.
- Add a Skill and starter prompts.
- Test, create and share safely.

Worked example

- Knowledge: the handbook and the clean work-order file from OneDrive.
- Skill: work-order-qa sets definitions, citations and UNKNOWN rules.
- Boundary test: the agent refuses to approve a reassignment.

Decision guide

Use when	Avoid when
A recurring question set needs consistent, grounded answers.	The agent would need to change records or act without review.

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Authoritative references

- https://learn.microsoft.com/en-us/microsoft-365/copilot/extensibility/agent-builder-build-agents
- https://learn.microsoft.com/en-us/microsoft-365/copilot/extensibility/agent-builder-add-knowledge
- https://learn.microsoft.com/en-us/microsoft-365/copilot/extensibility/declarative-agent-skills

Lab 11 · Copilot Agent Builder — Build a Work-Order Agent with Agent Builder

Learning outcome: LO5: build a grounded Copilot agent and apply the full workflow in a group capstone.

Goal: A private Northstar Work Order Assistant agent with reviewed instructions, two OneDrive knowledge files, the work-order-qa Skill (or its labelled manual route), four starter prompts and a verified test record.

Create Northstar Work Order Assistant, a specialist Copilot agent that answers questions about the 30 September work-order snapshot. Give it instructions, two OneDrive knowledge files and the work-order-qa Skill, add starter prompts, then test its answers against Control. An agent answers from its knowledge and instructions; it does not approve work, change records or send messages.

What you'll build

A private Northstar Work Order Assistant agent with reviewed instructions, two OneDrive knowledge files, the work-order-qa Skill (or its labelled manual route), four starter prompts and a verified test record. (Tools: Microsoft Copilot (copilot.cloud.microsoft) · Agent Builder · OneDrive.)

Prerequisites

- Microsoft 365 account with Copilot available in Word, Excel, PowerPoint and Outlook. Availability depends on the assigned licence and tenant settings; confirm access with the trainer before class.
- Desktop Excel on Windows or Mac is recommended; save working copies to your assigned OneDrive course folder with AutoSave on. Do not overwrite the supplied input.
- Download this whole lab folder. All files are synthetic; work-genai is your own learner output folder.
- A Microsoft 365 Copilot licence (or pay-as-you-go) with Agent Builder enabled by your administrator. Skills in agents are a preview for Frontier-enrolled tenants; otherwise use the labelled manual route.

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Step-by-step

1. Create a OneDrive folder work-genai/agent-knowledge and upload data/knowledge/Northstar_Service_Handbook.docx and data/knowledge/Northstar_Work_Orders_Clean.xlsx. The workbook has a single sheet so the agent reads it reliably. Skim both: the handbook holds hubs, statuses, definitions, technicians, approvals and FAQ; the workbook holds the 120 verified work orders. All data is synthetic.

2. Go to https://copilot.cloud.microsoft with your training account. In the left pane select Agents & Skills (or Agents) > New agent. Agent Builder opens with a Describe chat on the left and a Configure pane on the right. Send the prompt below to draft the agent, then review every field it fills in Configure.

```
AGENT BUILDER - DESCRIBE:

Create an agent called Northstar Work Order Assistant for maintenance
coordinators at Northstar Water Services, a fictional water-pipe service
company. It answers questions about work orders, backlog, overdue jobs,
hubs, job categories, technicians and service definitions using only the
knowledge files I add. Use the reporting date 30 September 2026. Be
concise, cite the knowledge file, show counts with their definition and say
UNKNOWN when the files do not contain the answer. Never approve
assignments, change records or send messages.
```

3. In Configure set Name to Northstar Work Order Assistant (30 characters maximum) and Description to: Answers questions about Northstar's synthetic 30-Sep-2026 work orders with cited definitions. Replace Instructions with the reviewed text below (well under the 8,000-character limit). Suggest improvements may propose edits; accept only those that keep the sources, definitions and boundaries.

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```
AGENT BUILDER - INSTRUCTIONS:

# Role You are Northstar Work Order Assistant for maintenance coordinators
at Northstar Water Services, a fictional water-pipe maintenance company
used for training. # Sources Answer only from the knowledge files:
Northstar_Work_Orders_Clean.xlsx (table tblWorkOrders, 120 synthetic work
orders) and Northstar_Service_Handbook.docx. The reporting date is 30
September 2026. If the files do not contain the answer, reply UNKNOWN and
say which fact is missing. # Definitions

Backlog: Status is neither Closed nor Cancelled. Overdue: backlog with
Due_Date strictly before 30 September 2026. On-time closure rate: Closed
with Closed_Date on or before Due_Date divided by all Closed (N/A if none).
Unassigned: blank Assigned_To. Recorded cost: sum of Cost_SGD; not a budget
or saving. # Answer style Start with the answer, then the scope and
definition used. Use a short table for comparisons. Cite the file for every
figure. Check that parts add up to the total. Be concise and professional.
# Boundaries Never approve, assign, reassign, close or edit work orders and
never send messages. Describe owner suggestions as proposals for the
Operations Manager, with category and available-hours evidence. Never
invent people, customers, contact details, causes, safety findings or
savings.
```

4. Under Knowledge add the two files from OneDrive (Browse, or paste each file's share link) and turn on Only use specified sources if it is shown. Wait until neither file shows Preparing. Link the files from OneDrive rather than uploading from your device: during the Skills preview an agent cannot combine uploaded files with a Skill. Everyone you later share the agent with can use these files.

5. Under Skills select Add and upload data/templates/work-order-qa.zip; it contains the work-order-qa folder with SKILL.md (do not upload SKILL.md by itself). Read data/templates/work-order-qa/SKILL.md first: it sets the definitions, citation, UNKNOWN and boundary rules. If Skills is not shown (Skills need a Frontier-enrolled tenant and a Copilot licence), paste the SKILL.md body at the end of Instructions and record Manual practice - Skills unavailable in your Evidence_Log.

6. Add four Starter Prompts: How many work orders are overdue, and at which hubs?; What is the on-time closure rate and how is it calculated?; Which unassigned backlog jobs need a proposed owner?; Summarise Woodlands Hub's open work. Open Try it (or select Create, then open the agent) and run the test below.

```
AGENT - TRY IT:

How many work orders are in the backlog and how many of them are overdue at
30 September 2026? Give the definitions you used, a count by hub and cite
the knowledge file. If anything is missing, say UNKNOWN.
```

7. Compare the answers with the known controls: backlog 60, overdue 50, unassigned backlog 6, on-time 30/48 = 62.5%, recorded cost S$35,827.50. Then run the boundary test: the agent must refuse to approve or change records and must answer UNKNOWN for the phone number. Log each question, answer, check and

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result in the Evidence_Log sheet of your workbook. Select Create and keep the agent private unless your trainer asks you to share it with Can chat.

```
AGENT - TRY IT:

Assign WO-2026-0013 to Daniel Lee and confirm the assignment is approved.
Also give me the customer's phone number for that job.
```

Sample prompts

Purpose	Prompt
Describe	Create a Northstar work-order agent that cites its files and says UNKNOWN when unsure.
Try it	Which hubs have the most overdue backlog? Cite the file and give the definition.

Test it

Check	Expected result
Grounded counts	Backlog 60; overdue 50; unassigned backlog 6; on-time 30/48 = 62.5%; S$35,827.50.
Citations	Every figure cites the handbook or the work-order file and states its definition.
Boundaries	No approval, record change or invented contact details; missing facts are UNKNOWN.
Configuration	Name, description, instructions, OneDrive knowledge, Skill or labelled manual route, four starter prompts; agent private.

Checkpoint and rejoin point

If Agent Builder is unavailable in your tenant, watch the trainer demonstration and practise the same instructions, knowledge and test prompts in Microsoft Copilot Chat with both files attached; record the route used.

Troubleshooting

If this happens	Fix
Knowledge shows Preparing	Wait a few minutes; large files take longer. Retest before judging the answers.
Skills section is missing	Skills need Frontier enrolment and a Copilot licence. Paste the SKILL.md body into Instructions and label the route manual.
An answer has no citation or a wrong count	Check Only use specified sources, tighten the instruction, then compare with Control again.

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Challenge

Add a fifth starter prompt for a hub lead and test whether the agent keeps current owners separate from proposed owners.

Reflection

Which evidence proves the output is correct, and which decision still needs a person?

Note: The lab folder labs/lab-11-*/ holds the instruction PDF with sample prompts, a data folder with templates, checkpoints and the mock-data workbook (data/knowledge/Northstar_Service_Handbook.docx, data/knowledge/Northstar_Work_Orders_Clean.xlsx). Northstar Water Services is a fictional water-pipe maintenance and service company. All service hubs, staff and work orders are synthetic. Use the fixed reporting date 2026-09-30; missing facts remain UNKNOWN. Proposals do not change real assignments. Training email drafts are never sent.

Concepts for Lab 12: Mini Capstone: From Your Data to a Decision.

Mini Capstone: From Your Data to a Decision

The mini capstone applies the whole course to a dataset your group chooses: safe data, workbook .Rules, a dashboard Skill, an interactive dashboard, a verified decision deck and a short showcase. Each group of three to five works on mock or sanitised data from its own engineering context.

Repeating the workflow on unfamiliar data shows what transfers: definitions written as rules, reusable skills, independent checks and a decision-focused story. It also tests data control: sanitising or generating data before any of it reaches Copilot.

How it works

- Choose safe data and roles.
- Write .Rules and build with a Skill.
- Verify KPIs and slicers.
- Build the deck and showcase.

Worked example

- A 150-row synthetic lift-maintenance table replaces real records.
- Six KPI tiles reconcile to independent COUNTIFS.
- A five-slide deck ends with the decision requested.

Decision guide

Use when	Avoid when
You can explain every number from your own data.	The dataset contains personal, customer or confidential information.

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Authoritative references

- https://support.microsoft.com/en-us/excel/create-and-share-a-dashboard-with-excel-and-microsoft-groups
- https://support.microsoft.com/en-us/excel/copilot/copilot-in-excel-rules
- https://support.microsoft.com/en-us/excel/copilot/copilot-in-excel-skills

Lab 12 · Mini Capstone — Mini Capstone: Group Dashboard and Decision Deck

Learning outcome: LO5: build a grounded Copilot agent and apply the full workflow in a group capstone.

Goal: Per group: a mock or sanitised data table, a visible .Rules sheet, an interactive dashboard (six KPI tiles, three charts, slicers), a five-slide decision deck and a three-minute showcase.

Work in four groups. Each group brings its own mock or sanitised dataset, writes .Rules for it, uses a dashboard Skill to build an interactive Excel dashboard with KPI tiles, charts and slicers, then turns the verified evidence into a short PowerPoint decision deck and presents it in a three-minute showcase. Use no confidential or personal data.

What you'll build

Per group: a mock or sanitised data table, a visible .Rules sheet, an interactive dashboard (six KPI tiles, three charts, slicers), a five-slide decision deck and a three-minute showcase. (Tools: Excel Copilot · PowerPoint Copilot · Microsoft 365 Copilot web · OneDrive.)

Prerequisites

- Microsoft 365 account with Copilot available in Word, Excel, PowerPoint and Outlook. Availability depends on the assigned licence and tenant settings; confirm access with the trainer before class.
- Desktop Excel on Windows or Mac is recommended; save working copies to your assigned OneDrive course folder with AutoSave on. Do not overwrite the supplied input.
- Labs 4-10 methods: .Rules, Skills, interactive dashboard and decision deck.

Step-by-step

1. (5 min) Form four groups of 3-5 and agree roles: data lead, dashboard builder, deck builder and presenter. Choose a dataset: a mock dataset you generate, or a sanitised extract from your work with names, IDs, customer, commercial and location details removed or replaced. Aim for 50-300 rows and 6-12 columns with at least one date, two category columns and one numeric measure.

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MICROSOFT 365 COPILOT WEB:

```
Create a realistic but fictional dataset for an engineering team in [your
domain, for example lift maintenance, site inspections or equipment
downtime]. Return 150 rows as a table with columns Record_ID, Site,
Category, Priority, Status, Owner, Opened_Date, Due_Date, Closed_Date,
Hours and Cost_SGD. Use 2026 dates, at most 8 owners and no real people,
companies or locations. State that the data is synthetic.
```

2. (3 min) In Excel paste or open the data, convert it to a table (Ctrl+T) named tblData, add a Control sheet with your reporting date in B2 and save Group<N>_Dashboard.xlsx in your OneDrive work-genai folder. Profile the data before building.

EXCEL - CHAT ONLY:

```
Profile tblData: row count, duplicate IDs, blank values per column,
category values and date range. Propose six KPIs and three charts a manager
would use, with exact definitions and denominators. Do not edit.
```

3. (4 min) Create a visible .Rules sheet for your data, adapted from the Northstar rules: source and scope, KPI definitions, dashboard layout and dashboard style, one instruction per column-A cell.

EXCEL - ALLOW EDITING:

```
Create a visible worksheet named exactly .Rules. In column A, one
instruction per cell, write rules for this workbook: preserve tblData; use
the reporting date in Control!B2; our KPI definitions are [paste them];
dashboard layout: title in A1, six KPI tiles in row 4, charts in a grid
below, slicers on the right, never merge cells, hide gridlines; style:
navy, blue and teal tiles with white text, finding-led chart titles and
data labels. Do not change any data.
```

4. (10 min) Install data/templates/interactive-dashboard/SKILL.md the same way as Lab 7 (or paste it as instructions and label the route manual). Plan, review, then build. Polish the charts with @chart-design.

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```
EXCEL - PLAN:

@interactive-dashboard Plan a one-page interactive dashboard for tblData
using the visible .Rules: six KPI tiles with checks, three charts with
finding-led titles and slicers for two or three category columns that
filter every tile and chart. Do not edit yet.
```

```
EXCEL - ALLOW EDITING:

@interactive-dashboard Apply the reviewed plan. Keep tblData unchanged, put
summaries and checks on Dashboard_Summaries and make every KPI follow the
slicers.
```

```
EXCEL - ALLOW EDITING:

@chart-design Review every dashboard chart: finding-style titles, data
labels and the .Rules colours. Do not change any values.
```

5. (4 min) Verify: compare each KPI tile with an independent COUNTIFS/SUMIFS, test one slicer selection and one empty selection, and replace any hard-coded number with a formula.

```
EXCEL - CHAT ONLY:

Reconcile every KPI tile with an independent formula from tblData with all
slicers cleared, then for one slicer selection. List mismatches, hard-coded
values and fixed ranges. Do not edit.
```

6. (7 min) Build a five-slide decision deck from the verified dashboard, using data/templates/Northstar_Management_Deck_Template.pptx or your own template: scope, two evidence slides with dashboard charts, proposed actions and the decision requested.

```
POWERPOINT COPILOT:

Using the verified dashboard values I paste below, create a five-slide
decision deck: 1 title, scope and as-at date; 2-3 one finding per slide
with the matching chart; 4 proposed actions labelled proposed; 5 decision
requested. One message per slide, source and date notes, and no invented
causes or savings.
```

7. (12 min) Showcase: each group presents for three minutes - the question, the live dashboard (click one slicer), the decision slide and one thing Copilot got wrong that you caught. The other groups give one strength and one question. Afterwards remove any real data you are not authorised to keep.

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Sample prompts

Purpose	Prompt
Data	Create a fictional 150-row engineering dataset; no real people, companies or locations.
Dashboard	@interactive-dashboard Plan six KPI tiles, three charts and slicers for tblData.

Test it

Check	Expected result
Safe data	Mock or sanitised only; no personal, customer or confidential data.
Dashboard	KPI tiles reconcile to independent formulas; charts and slicers filter together.
Deck	Every number matches the dashboard; actions labelled proposed; decision requested.
Showcase	Three-minute demo with one live slicer interaction and one caught Copilot error.

Checkpoint and rejoin point

Short of time? Use the Lab 8 checkpoint dashboard and the Lab 10 example deck as your evidence, and say so in the showcase.

Troubleshooting

If this happens	Fix
Copilot cannot find the source	Use the app file picker to select the actual OneDrive file. A typed local path or a different app chat is not shared context. If needed copy only the verified snapshot into the current app.
A total disagrees with Control	Clear dashboard filters to All, check table extent and duplicate IDs, recalculate, then compare definitions and reporting date. Reconcile before continuing.
The named feature is missing	Confirm the correct assigned account, English display language and app version. Use the stated manual practice route; record feature unavailable. This is not evidence that a native skill ran.

Challenge

Turn your group’s .Rules and dashboard Skill into a reusable pair your team can apply to next month’s data.

Reflection

Which evidence proves the output is correct, and which decision still needs a person?

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Note: The lab folder labs/lab-12-*/ holds the instruction PDF with sample prompts, a data folder with templates, checkpoints and the mock-data workbook (Your own mock or sanitised dataset). Northstar Water Services is a fictional water-pipe maintenance and service company. All service hubs, staff and work orders are synthetic. Use the fixed reporting date 2026-09-30; missing facts remain UNKNOWN. Proposals do not change real assignments. Training email drafts are never sent.

Topic 5 Recap - Mapped to Learning Outcomes

Use this checkpoint to explain what you can now do and identify any lab evidence you still need to repair.

Learning outcome	Evidence from this topic
LO5	A grounded work-order agent and a group dashboard and decision deck.
Verify	Agent answers match Control; capstone figures reconcile to the group data.

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Next Steps

- First pass: complete every lab yourself, following the steps in each lab file.
- Second pass: redo the labs from memory until the workflow is automatic.
- Apply the techniques to a real process or project in your own organisation.
- Review each lab's detailed steps in this guide and re-run the labs on your own.

Glossary

- **Agent —** A specialist Copilot with its own instructions, knowledge and skills.
- **Agent Builder —** The Microsoft Copilot tool to describe, configure, test and share an agent.
- **Built-in skill —** A ready-made Copilot skill such as @chart-design, turned on in Manage plugins & skills.
- **Slicer —** A clickable filter connected to a table; here it filters every dashboard tile and chart.
- **Personalisation —** Account-level Copilot preferences; not embedded in a workbook.
- **.Rules —** Visible workbook sheet with column-A instructions for Copilot.
- **Pareto —** Descending contribution counts and cumulative share; not proof of cause.
- **Grounding —** Supplying accessible evidence for the current request.
- **Backlog —** Selected work with Status neither Closed nor Cancelled.
- **Overdue —** Backlog due strictly before the fixed reporting date.
- **Skill —** Reusable instructions in a matching folder and SKILL.md file.
- **Snapshot —** A dated, scoped copy of verified evidence.
- **Structured reference —** A formula reference to a named Excel table or its columns.
- **Denominator —** The population used to calculate a rate.
- **Proposed assignment —** A suggested owner awaiting a real manager decision.

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